

Forward KL Can Be Better: Normalizing Flow Boltzmann Sampling with Log-Ratio Variation

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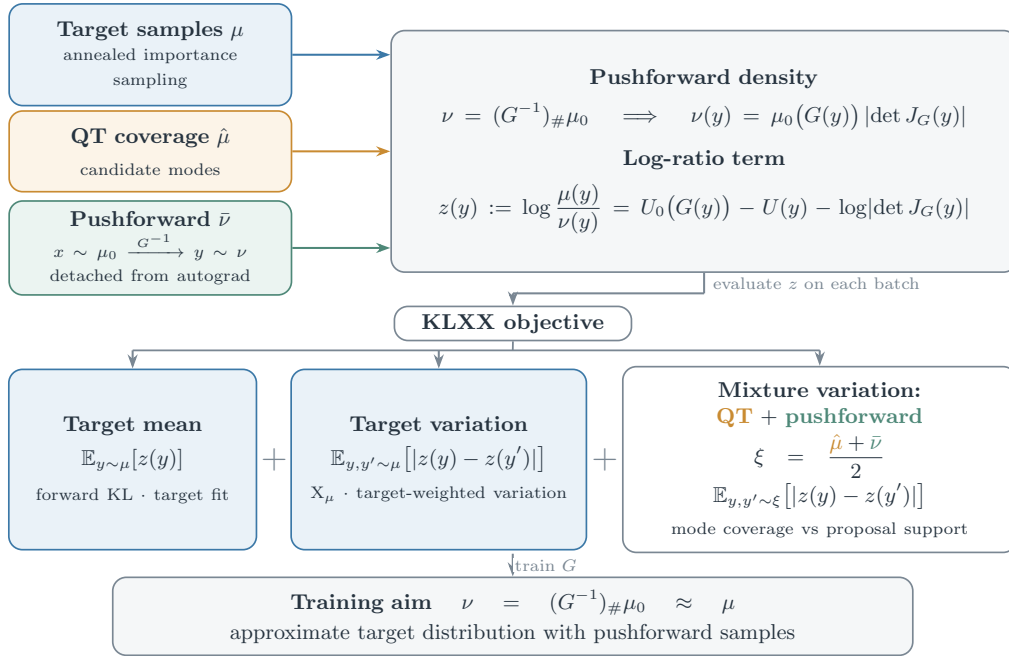
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Abstract

Normalizing flow Boltzmann generators retain a tractable proposal density, but training with forward KL depends on target samples that may be biased or omit modes. A flow can then miss target mass while its observed importance weights give a high effective sample size. We introduce log-ratio variation, a pairwise regularizer of the exact target-to-proposal log-density ratio, and use it to define KLXX. Its three weighting measures address target fitting, candidate mode coverage, and proposal support control without changing the exact target minimizer. We derive an exponential Fisher–Rao bound and a fixed-surrogate error bound for forward KL with the target-weighted term, then place KLXX in an adaptive-staging Boltzmann generator with exact importance reweighting and sharpening of the regularized molecular potential. In the reported model and molecular tests, KLXX improves mode-coverage or stagewise diagnostics, and selected molecular observables agree with independent molecular simulations. Log-ratio variation thus supplies information that a sampled forward KL objective can omit.

KLXX: one log-ratio (KL), two log-ratio variations (XX)



1 Introduction

Drawing samples from a target distribution $\mu \propto \exp(-U)$ on $\mathbb{R}^d$ is a recurring problem in molecular simulation, Bayesian inference, and statistical physics (Frenkel and Smit, 2002; Robert and Casella, 2004; Landau and Binder, 2014). When μ is high-dimensional and multimodal, its mass splits among basins of U separated by barriers. Local sampling kernels, including Metropolis–Hastings (Metropolis et al., 1953; Hastings, 1970), Langevin dynamics (Langevin, 1908; Leimkuhler and Matthews, 2015), Nosé–Hoover dynamics (Nosé, 1984; Hoover, 1985), and Hamiltonian Monte Carlo (Duane et al., 1987), evolve from the current configuration and can therefore remain confined to one basin over accessible simulation times. When this occurs, finite-time estimates are biased toward the visited basin and give no direct account of the mass that was never reached.

Tempering and particle methods address this problem by connecting a tractable distribution to μ through intermediate distributions. Parallel tempering (PT) and replica exchange run copies of the system at several temperatures and exchange neighboring configurations, so a basin reached at high temperature can enter the low-temperature ensemble (Swendsen and Wang, 1986; Geyer, 1991; Hukushima and Nemoto, 1996). Annealed importance sampling (AIS) moves along a bridge of intermediate distributions while accumulating importance weights (Neal, 2001). Sequential Monte Carlo (SMC) uses a similar bridge but resamples and rejuvenates a weighted particle set at each level (Doucet et al., 2001; Del Moral et al., 2006). For a prescribed chemical process, umbrella sampling biases selected reaction coordinates to cross the relevant barriers and reweights the biased samples to recover a free energy profile (Torrie and Valleau, 1977; Kästner, 2011). Nested sampling instead rewrites the normalizing integral as a one-dimensional quadrature over a likelihood threshold (Skilling, 2006).

Recent work improves PT through three distinct mechanisms. Nonreversible swap schedules and tuned temperature grids accelerate round trips (Syed et al., 2022); variational-reference methods adapt the reference endpoint and its connecting path (Surjanovic et al., 2022); and neural transports improve communication between adjacent distributions with limited overlap (Zhang et al., 2025). All three strengthen communication between neighboring distributions and can transmit a discovered basin efficiently, but none removes the dependence on local exploration: a basin that no chain has visited cannot enter the target ensemble (Woodard et al., 2009; Chehab et al., 2025). A complementary strategy is to learn a global transport from an easily sampled reference to the target, thereby generating independent, nonlocal proposals.

Boltzmann generators implement this strategy with a learned transport map (Noé et al., 2019). Let $\mu_0 \propto \exp(-U_0)$ be an easily sampled reference distribution, usually Gaussian, and let $G : \mathbb{R}^d \rightarrow \mathbb{R}^d$ be an invertible normalizing flow (Rezende and Mohamed, 2015; Papamakarios et al., 2017; Kingma and Dhariwal, 2018). We use the target-to-source convention $G_{\#}\mu \approx \mu_0$. The inverse map then sends independent source draws toward the target, and their pushforward distribution is $\nu := (G^{-1})_{\#}\mu_0$. Its density is

$$\nu(y) = \mu_0(G(y)) |\det J_G(y)|, \quad (1)$$

where $J_G(y)$ is the Jacobian matrix of G at y . Importance weights proportional to μ/ν are therefore available on pushforward samples without fitting another density or tracing a stochastic path. These weights enable importance-reweighted estimates and define the effective sample size (ESS), a diagnostic of density agreement on the support reached by ν (Liu, 2001; Doucet et al., 2001; Elvira et al., 2022). The remaining challenge is that a high ESS certifies agreement only on the reached support: a target mode absent from the pushforward samples contributes no observed weight.

The choice of objective determines how this challenge enters flow training. The original Boltzmann generator (Noé et al., 2019) and its variational-inference predecessor (Rezende and Mohamed, 2015) minimize the reverse Kullback–Leibler (KL) divergence

$$(\text{reverse}) \quad \text{KL}(\nu \parallel \mu) = \int_{\mathbb{R}^d} \nu(x) \log \frac{\nu(x)}{\mu(x)} dx, \quad (2)$$

which needs samples only from ν and evaluations of U . Its empirical gradient is therefore accessible, but the divergence is mode-seeking (Wainwright and Jordan, 2008): a mode absent from ν contributes no samples and hence no gradient. A flow can then collapse onto a subset of the target modes (Wu et al., 2020; Felardos et al., 2023). Existing approaches interleave Markov chain Monte Carlo (MCMC) kernels with deterministic layers, encode target symmetries in the architecture, reduce gradient variance near the optimum, or anneal

from a smoother potential (Wu et al., 2020; Köhler et al., 2020; Vaitl et al., 2022; Schopmans and Friederich, 2025). These changes can improve exploration, but their success still depends on whether the resulting sampler reaches each relevant mode.

The opposing objective minimizes the forward KL,

$$\text{(forward)} \quad \text{KL}(\mu\|\nu) = \int_{\mathbb{R}^d} \mu(x) \log \frac{\mu(x)}{\nu(x)} dx, \quad (3)$$

which is mass-covering for exact distributions: it diverges if ν vanishes where μ has positive mass (Wainwright and Jordan, 2008). Its empirical form, however, needs samples from the target distribution μ being learned. Existing work resolves this circularity in several ways. Maximum likelihood uses configurations from a reference thermodynamic state (Wirnsberger et al., 2020); adaptive flow samplers train on a running Markov chain and use the flow as a global proposal (Gabri   et al., 2022); molecular flows can fit a molecular-dynamics trajectory (Klein et al., 2023); and transferable architectures amortize coverage across related molecules (Klein and No  , 2024). Persistent chains and coupled-target regression provide other approximations to the missing target expectation (Naesseth et al., 2020; Rehman et al., 2026). Because an empirical loss reflects only the regions represented by its sample source, the mass-covering property of the exact divergence does not prevent empirical mode omission. This tension motivates an objective that retains forward KL while using samples from more than one distribution.

The sampled forward KL leaves three distinct gaps. First, a mode absent from its target surrogate contributes no training points. If the flow also misses that mode, the observed weights can be nearly uniform on the reached support, so the ESS can be close to one even though target mass is absent; we call this a *fake ESS* (No   et al., 2019; Midgley et al., 2022). Second, a finite or biased target surrogate distorts the sampled objective, and the fitted flow can inherit that error. Third, target samples give little information where the proposal places excess mass, allowing leakage between modes or into target-poor regions. Addressing one of these failures does not by itself resolve the other two.

To address these gaps, we propose KLXX, a forward KL loss augmented by log-ratio variations; the graphical abstract summarizes it, and Equation (8) gives its full form. KLXX preserves the tractable target-to-proposal density ratio and exact target minimizer while separating target fitting, candidate mode coverage, and proposal support control. The first two terms of (8), the sampled forward KL and the target-weighted variation X_μ , both draw on the target surrogate supplied by annealing. The new third term mixes samples from quench and temper (QT) with pushforward samples: QT contributes candidate modes missing from the surrogate, while pushforward samples expose proposal support and leakage. QT guides coverage rather than estimating physical mode weights or guaranteeing complete discovery.

The closest related method is Flow Annealed Importance Sampling Bootstrap (FAB), which couples AIS and flow training but replaces the forward KL by the α -divergence ($\alpha = 2$) (Midgley et al., 2022). Its integrand is proportional to μ^2/ν , and its exact minimizer also minimizes importance-weight variance. FAB targets μ^2/ν with AIS and reuses those samples through a prioritized replay buffer, which must be maintained across training. Related methods insert flows into SMC, anneal toward low-temperature targets, add an inference-time SMC correction, or connect disconnected regions (Arbel et al., 2021; Matthews et al., 2022; Schopmans and Friederich, 2025; Tan et al., 2025; von Klitzing et al., 2026). KLXX instead retains the forward KL and adds nonnegative log-ratio variations. Its additional cost is the QT construction and the variation evaluations, including the extra mixture batch; it uses no replay buffer. FAB’s 2-divergence gives a stronger tail penalty, whereas KLXX obtains coverage information from QT. Section 3.2 compares their sampling mechanisms.

Centered log-ratio penalties also arise in log-variance objectives for learned diffusion samplers (Richter and Berner, 2024) and in log-dispersion regularization (LDR) objectives for Boltzmann generators (Schopmans et al., 2026). Although the present work was developed independently of LDR, these connections place our construction within a broader family of log-ratio regularizers. Our target-weighted term differs by pairing the forward KL integrand at two independent target points, which yields the double-integral variation in (4). KLXX further evaluates this variation under a mixture of QT and pushforward samples.

A separate line trains a stochastic process rather than a bijection. Controlled stochastic differential equations, denoising diffusions, and flow-matching models learn a drift or velocity that transports the reference toward μ (Zhang and Chen, 2022; Vargas et al., 2023; Berner et al., 2024; N  sken and Richter, 2021; Richter and Berner, 2024). Energy-only variants such as iterated denoising energy matching (iDEM) alternate a

sampling loop with a score- or velocity-fitting loop and a replay buffer (Akhound-Sadegh et al., 2024; Woo and Ahn, 2024). Learned nonequilibrium transport augments AIS with a trainable drift, and later work extends this approach to larger many-body systems (Albergo and Vanden-Eijnden, 2025; Havens et al., 2025; Akhound-Sadegh et al., 2025).

These stochastic transports can be more expressive than the invertible architectures used here. Their terminal density, however, often requires per-sample ordinary differential equation (ODE) integration (He et al., 2025). We retain a bijection because the exact proposal density, importance ratio, ESS, and reweighting are central to the objective and final estimator. Table 1 summarizes these distinctions.

method	objective	training mechanism
reverse KL	$\text{KL}(\nu\ \mu)$	pushforward samples; no target sampler
forward KL	$\text{KL}(\mu\ \nu)$	annealed target surrogate in this work
FAB	2-divergence (μ^2/ν)	AIS on μ^2/ν ; MCMC differentiates ν ; replay buffer
LDR	$\text{KL}(\mu\ \nu) + \mathbb{E}_\mu[z - \mathbb{E}_\mu z ^p]$	fixed energy-labeled data; no extra pushforward samples
iDEM	energy matching	iterative sampler; stochastic transport; replay buffer
KLXX	$\text{KL}(\mu\ \nu) + X_\mu + X_{(\hat{\mu} + \bar{\nu})/2}$	target annealing, QT, and detached pushforward samples no MCMC derivative of ν or replay buffer

Table 1: Objectives and training mechanisms for the methods discussed above. Here $z = \log(\mu/\nu)$; the LDR row gives its target-weighted form with $p = 1$ for LDR-L1 and $p = 2$ for LDR-L2. The forward KL row uses the annealed baseline in this paper but may instead use external target data. The table compares how samples enter each objective; it does not assume equal computational budgets.

We now turn from these method-level comparisons to the mathematical definition of KLXX. Define the tractable log-density ratio $z(x) := \log \frac{\mu(x)}{\nu(x)}$. For a nonnegative density ω on $\mathbb{R}^d$, define the *log-ratio variation*

$$X_\omega(\mu\|\nu) = \iint_{\mathbb{R}^d \times \mathbb{R}^d} \omega(x) \omega(y) \left| \log \frac{\mu(x)}{\nu(x)} - \log \frac{\mu(y)}{\nu(y)} \right| dx dy, \quad (4)$$

which measures how far z is from a constant on the region weighted by ω . For $\omega = \mu$, X_μ averages $|z(x) - z(y)|$ over independent target points, whereas LDR-L1 uses the centered penalty $\mathbb{E}_\mu|z - \mathbb{E}_\mu z|$. Jensen’s inequality and the triangle inequality give $\mathbb{E}_\mu|z - \mathbb{E}_\mu z| \leq X_\mu(\mu\|\nu) \leq 2\mathbb{E}_\mu|z - \mathbb{E}_\mu z|$. Thus, at the same coefficient, $\text{KL} + X_\mu$ and LDR-L1 are comparable up to a factor of two. This suggests similar training performance, although the objectives are not equivalent. $\text{KL} + X_\mu$ is also not equivalent to LDR-L2: its target-weighted variation is pairwise L^1 , and its new mixture term uses QT and pushforward samples. Since X_ω is nonnegative and vanishes at $\nu = \mu$, adding it to the exact forward KL preserves the target as a minimizer.

The target distribution μ supplies samples for both the forward KL and X_μ . The wide-coverage measure $\hat{\mu}$ supplies candidate-mode samples, while $\bar{\nu}$ is exactly the pushforward distribution ν of the current proposal, detached from the differentiation graph. In other words, $\bar{\nu}$ is a *stop-gradient* copy of ν : it supplies the same samples, but they are held fixed when the loss is differentiated. The combined loss is

$$\mathcal{L}[\nu] = \text{KL}(\mu\|\nu) + \lambda X_\mu(\mu\|\nu) + X_{\alpha\hat{\mu} + \beta\bar{\nu}}(\mu\|\nu), \quad (5)$$

where $\lambda, \alpha, \beta \geq 0$. For simplicity, we assume $\alpha + \beta = 1$ in this section; Section 3.1 allows arbitrary nonnegative mixture weights. Under this convention, the final term is the log-ratio variation under the mixture distribution $\alpha\hat{\mu} + \beta\bar{\nu}$. Its pairwise expectation includes pairs drawn within either distribution and pairs with one sample from each distribution.

To rewrite the loss (5) in a tractable map-level form, use $\nu = (G^{-1})_\# \mu_0$, which implies the exact identity $G_\# \nu = \mu_0$. Combining (1) with $\mu(y) = \exp(-U(y))$ and $\mu_0(x) = \exp(-U_0(x))$ gives the following expression for the log-density ratio defined above:

$$z(y) = \log \frac{\mu(y)}{\nu(y)} = U_0(G(y)) - U(y) - \log |\det J_G(y)|. \quad (6)$$

Substituting (6) into (5) yields

$$\mathcal{L}[G] = \underbrace{\mathbb{E}_{y \sim \mu} [z(y)]}_{\text{KL}(\mu \parallel \nu)} + \underbrace{\lambda \mathbb{E}_{y, y' \sim \mu} [|z(y) - z(y')|]}_{X_\mu(\mu \parallel \nu)} + \underbrace{\mathbb{E}_{y, y' \sim \alpha \hat{\mu} + \beta \bar{\nu}} [|z(y) - z(y')|]}_{X_{\alpha \hat{\mu} + \beta \bar{\nu}}(\mu \parallel \nu)}. \quad (7)$$

KLXX is formally defined by the loss (7) with the default choice $(\lambda, \alpha, \beta) = (1, \frac{1}{2}, \frac{1}{2})$, and explicitly reads

$$\text{KLXX}[G] = \underbrace{\mathbb{E}_{y \sim \mu} [z(y)]}_{\text{KL}(\mu \parallel \nu)} + \underbrace{\mathbb{E}_{y, y' \sim \mu} [|z(y) - z(y')|]}_{X_\mu(\mu \parallel \nu)} + \underbrace{\mathbb{E}_{y, y' \sim (\hat{\mu} + \bar{\nu})/2} [|z(y) - z(y')|]}_{X_{(\hat{\mu} + \bar{\nu})/2}(\mu \parallel \nu)}. \quad (8)$$

With equal target and mixture batch sizes, KLXX evaluates about twice as many samples per step as forward KL and also requires the QT construction.

Our contributions are as follows. First, we introduce the KLXX objective, whose new component mixes QT and pushforward samples. Pairwise differences remove the unknown target normalizer, and the exact target loss retains $\nu = \mu$ as a minimizer. The target-weighted variation reuses the forward KL batch, while the mixture supplies candidate-mode and proposal support information absent from that batch.

Second, we analyze forward KL with X_μ in Fisher–Rao geometry. The regularizer contributes a nonpositive term to the instantaneous KL dissipation in (12). For a fixed biased target surrogate and a regularization strength below the threshold of Theorem 1, it also subtracts a nonnegative discrepancy term from the stationary-error bound. These results do not cover the full KLXX loss, the changing empirical surrogate used in training, or discrete neural-network optimization.

Third, we build an adaptive-staging Boltzmann generator without a replay buffer. Annealing and QT generate the required samples on the fly, and the MCMC transitions use only derivatives of U without differentiating the flow density. Each stage compares its trained proposal with the identity on the validation set, retains the proposal with higher ESS, and shrinks rejected stage increments. For molecular potentials, an additional continuation sharpens a regularized interaction core toward the desired endpoint.

Fourth, we demonstrate that QT can reach target basins outside the regions covered by the source and annealed samples. In the Himmelblau and Sparse tests, the two objectives trained without QT miss remote wells, while both QT-based losses recover every mode; mixing QT and pushforward samples also reduces visible intermodal leakage (Figures 3 and 4). We therefore report ESS together with coverage or a problem-specific mode diagnostic, since uniform weights on reached modes do not imply complete coverage. The clock comparison uses shared stage schedules, whereas the molecular stage schedules are method-specific and single-seed; the results support this mechanism without establishing universal or fixed-compute dominance over forward KL.

Fifth, we develop the Python packages `jflows` and `jflows-md` for normalizing flow Boltzmann sampling with molecular potentials. With a JAX backend and OpenXLA compilation, the reported 22-atom alanine dipeptide run completes all ten accepted training stages in 72.7 minutes of total persisted stage time on one NVIDIA GeForce RTX 5090, without an external target dataset or a prescribed conformational landscape.

Organization. Section 2 analyzes convergence and fixed-surrogate error for $\text{KL} + X_\mu$ in Fisher–Rao geometry; Appendix A gives the proofs. Sections 3 and 4 define the full KLXX construction and propose the adaptive-staging Boltzmann generator. Sections 5 and 6 report the model and molecular results, respectively, and Section 7 concludes.

2 Fisher–Rao gradient flow analysis

For simplicity, this section analyzes only forward KL with the target-weighted regularizer λX_μ as a density-space flow. Starting from the full loss (5), we omit the mixture term weighted by $\hat{\mu}$ and $\bar{\nu}$ and consider

$$\mathcal{L}[\nu] = \text{KL}(\mu \parallel \nu) + \lambda X_\mu(\mu \parallel \nu), \quad (9)$$

with fixed $\lambda > 0$. This restriction isolates the nonlocal rank correlation induced by X_μ from the additional mixture term built from QT and detached pushforward samples. It does not cover the full KLXX objective or its discrete neural-network optimization.

The metric determines the resulting density dynamics. In the Wasserstein-2 geometry the forward KL is not displacement convex, so the Otto–Villani (Otto and Villani, 2000) and Bakry–Émery (Bakry et al., 2014) arguments used for the reverse KL do not apply. We instead work in the Fisher–Rao (FR) geometry. We set $\text{sgn}(0) = 0$ and assume throughout that the densities are positive and sufficiently regular for the gradient calculations. Writing the ν -mean as $\langle \cdot \rangle_\nu := \int(\cdot) \nu dx$, the FR gradient of a smooth functional $\mathcal{F}$ and the gradient flow for $\mathcal{L}$ are

$$\text{grad}^{\text{FR}} \mathcal{F}(\nu) = \nu \left(\frac{\delta \mathcal{F}}{\delta \nu} - \left\langle \frac{\delta \mathcal{F}}{\delta \nu} \right\rangle_\nu \right), \quad \partial_t \nu_t = -\text{grad}^{\text{FR}} \mathcal{L}(\nu_t), \quad (10)$$

and the mean-subtraction makes $\int \text{grad}^{\text{FR}} \mathcal{F} = 0$, so the flow conserves probability mass.

For the loss (9), the gradient flow has the explicit form

$$\partial_t \nu_t(x) = \mu(x) - \nu_t(x) + 2\lambda \mu(x) S_t(x), \quad \text{where } S_t(x) := \mathbb{E}_{y \sim \mu} \left[\text{sgn} \left(\frac{\mu(x)}{\nu_t(x)} - \frac{\mu(y)}{\nu_t(y)} \right) \right]. \quad (11)$$

The local term $\mu(x) - \nu_t(x)$ is the contribution of forward KL to the FR gradient flow, while S_t is the nonlocal rank correlation: the update at x depends on how $\mu(x)/\nu_t(x)$ compares with the full distribution of importance weights under μ . Since $\mathbb{E}_\mu[S_t] = 0$, this term preserves total mass. Appendix A.1 derives (11).

Differentiating $\text{KL}(\mu \parallel \nu_t)$ along this flow, and reducing the X_μ term by a rank correlation identity, gives the dissipation

$$\frac{d}{dt} \text{KL}(\mu \parallel \nu_t) = -\chi^2(\mu \parallel \nu_t) - \lambda \mathbb{E}_{x, y \sim \mu} \left[\left| \frac{\mu(x)}{\nu_t(x)} - \frac{\mu(y)}{\nu_t(y)} \right| \right], \quad (12)$$

where $\chi^2(\mu \parallel \nu) := \int (\mu/\nu - 1)^2 \nu dx$ is the chi-squared divergence. The regularizer contributes a nonpositive term to the instantaneous KL dissipation. Since $\chi^2(\mu \parallel \nu_t) \geq \text{KL}(\mu \parallel \nu_t)$, Grönwall’s lemma yields

$$\text{KL}(\mu \parallel \nu_t) \leq e^{-t} \text{KL}(\mu \parallel \nu_0), \quad t \geq 0, \quad (13)$$

without assuming log-concavity, a spectral gap, or a growth condition on μ . This density-space statement does not imply that forward KL or KLXX trains faster with a discrete optimizer. It also excludes the cost of producing target samples. The added term in (12) describes how X_μ changes the density flow, not a universal transient speedup.

The convergence result assumes that the outer expectation is taken under the exact target. To isolate surrogate bias, fix a smooth density $\tilde{\mu}$ and replace μ by $\tilde{\mu}$ only in the outer integrals while retaining the exact pointwise log-ratio. This defines

$$\tilde{\mathcal{L}}[\nu] = \mathbb{E}_{x \sim \tilde{\mu}} \left[\log \frac{\mu(x)}{\nu(x)} \right] + \lambda \mathbb{E}_{x, y \sim \tilde{\mu}} \left[\left| \log \frac{\mu(x)}{\nu(x)} - \log \frac{\mu(y)}{\nu(y)} \right| \right]. \quad (14)$$

We assume that the densities are positive on their common domain and that the displayed KL, chi-squared, and pair integrals are finite. At $\lambda = 0$, the stationary density is $\nu^* = \tilde{\mu}$, so the residual error is $\text{KL}(\mu \parallel \tilde{\mu})$. The next result shows when X_μ improves this stationary-error bound. It is an idealization of one detached surrogate, not a theorem for the changing empirical batches used by the trainer.

Theorem 1 *Assume μ is absolutely continuous with respect to $\tilde{\mu}$ and $\lambda \in [0, \frac{1}{2})$. Then every stationary density ν^* of the gradient flow for the fixed-surrogate loss $\tilde{\mathcal{L}}$ in (14) satisfies*

$$\text{KL}(\mu \parallel \nu^*) \leq \text{KL}(\mu \parallel \tilde{\mu}) - \lambda \iint_{\mathbb{R}^d \times \mathbb{R}^d} \tilde{\mu}(x) \tilde{\mu}(y) \left| \frac{\mu(x)}{\nu^*(x)} - \frac{\mu(y)}{\nu^*(y)} \right| dx dy. \quad (15)$$

The subtracted term is nonnegative and is positive unless the displayed importance ratio is constant on the support of $\tilde{\mu}$. Within the theorem’s assumptions, the rank correlation induced by the second term in (14) raises ν^* in comparatively underweighted regions and lowers it elsewhere. The experiments use $\lambda = 1$, outside the sufficient range $\lambda < \frac{1}{2}$, so the theorem does not certify their finite-training behavior. Appendix A gives the variation, rank correlation identity, and proofs.

3 Flow training with log-ratio variation

This section turns the abstract loss of the introduction into a training algorithm for the flow map G . Let U_0 and U be the source and target potentials on $\mathbb{R}^d$, so that $\mu_0 \propto \exp(-U_0)$ is the samplable reference and $\mu \propto \exp(-U)$ is the target. We parameterize $G : \mathbb{R}^d \rightarrow \mathbb{R}^d$ so that $G_{\#}\mu \approx \mu_0$ and write $\nu := (G^{-1})_{\#}\mu_0$ for the induced approximation to μ . The loss itself evaluates only the native target-to-source map G and its log-Jacobian; it does not explicitly involve G^{-1} . The inverse map G^{-1} nevertheless generates pushforward samples used both to initialize AIS and to evaluate the $\bar{\nu}$ term; these G^{-1} evaluations are sampling-only, with the sampled coordinates detached from both the inverse-map path and their subsequent log-ratio evaluations differentiable through G . We first assemble the training loss and identify its sampling ingredients (Section 3.1), then supply them in turn: an AIS approximation of μ (Section 3.2) and the quench and temper construction of the wide-coverage measure $\hat{\mu}$ (Section 3.3). Section 3.4 gives the ESS and coverage diagnostics used to evaluate the trained flow.

3.1 Training loss with log-ratio variation

By change of variables, the pushforward density is $\nu(y) = \mu_0(G(y))|\det J_G(y)|$, so every term is a functional of the scalar field $z(y)$ in (6). The forward KL is its mean under the target, $\text{KL}(\mu\|\nu) = \mathbb{E}_{y \sim \mu}[z(y)]$. For the weighting measures used in training, we write $\omega(y)$ for the corresponding density; Equation (4) applies to any integrable nonnegative density, and empirical batches approximate its expectations by sample averages. The same formulas apply on a torus or a Euclidean-torus product with the corresponding standard reference volume. Set $s := \alpha + \beta$. For $s > 0$, let $\xi := (\alpha\hat{\mu} + \beta\bar{\nu})/s$ be the normalized mixture, with any zero-weight component omitted. Since $\alpha\hat{\mu} + \beta\bar{\nu} = s\xi$, the double integral in (4) produces the factor s^2 below. The map-level loss is

$$\mathcal{L}[G] = \underbrace{\mathbb{E}_{y \sim \mu}[z(y)]}_{\text{KL}(\mu\|\nu)} + \lambda \underbrace{\mathbb{E}_{y, y' \sim \mu}[|z(y) - z(y')|]}_{X_\mu(\mu\|\nu)} + s^2 \underbrace{\mathbb{E}_{y, y' \sim \xi}[|z(y) - z(y')|]}_{X_{\alpha\hat{\mu} + \beta\bar{\nu}}(\mu\|\nu)}. \quad (16)$$

When $s = 0$, the last term is omitted. Equation (16) is the map form of (5) after removing the introductory restriction $s = 1$; the factor s^2 follows from the double integral in (4) and is required when $\alpha + \beta \neq 1$. The default choice $\lambda = 1$ and $\alpha = \beta = \frac{1}{2}$ defines KLXX. The pushforward distribution $\nu = (G^{-1})_{\#}\mu_0$ depends on the trainable parameters of G . The barred measure $\bar{\nu}$ is the same distribution frozen at the current parameters: samples are drawn through G^{-1} and detached from that sampling path, but z at the sampled coordinates is still differentiated through the native map G .

The three sampling measures entering (16) address complementary failure modes of forward KL:

- μ : *target accuracy*. The term X_μ penalizes pairwise log-ratio variation within the target batch. Theorem 1 gives its density-space bound for $\lambda < \frac{1}{2}$; the finite-training effect at $\lambda = 1$ is tested empirically.
- $\hat{\mu}$: *candidate mode coverage*. QT provides log-ratio comparisons on basins represented in its wide-coverage set, including basins absent from the annealed batch. The melt scale m_e in Algorithm 2 controls how broadly QT searches.
- $\bar{\nu}$: *proposal support*. Pushforward samples evaluate the log-ratio where the current proposal places mass. The term is intended to reduce variation in target-poor regions; the experiments test whether this is accompanied by less visible leakage.

Adding these nonnegative terms to the exact target forward KL preserves $\nu = \mu$ as a minimizer. The terms change which samples enter the loss without changing that reference solution.

The forward KL and X_μ share one target batch. Evaluate $\mathbf{z} = (z(y_1), \dots, z(y_B))$, use $\text{mean}(\mathbf{z})$ for the forward KL, and use $\text{mean}(|\mathbf{z} - \mathbf{z}[\sigma]|)$ for X_μ , where σ is a uniformly random batch permutation. This estimator introduces no additional flow evaluation or differentiation because it reuses the same $\mathbf{z}$ and the same flow differentiation graph; the only added operations are the random permutation and pairwise difference. When $s > 0$, the last term evaluates one additional batch drawn from ξ , applies the same permuted-pair estimator, and multiplies it by s^2 . At the KLXX setting, each sample in this additional batch is drawn from $\hat{\mu}$ or $\bar{\nu}$ with probability $1/2$.

The loss draws on three measures, but $\bar{\nu}$ requires no construction beyond source draws, so at most two must be built. The $\bar{\nu}$ batch consists of pushforward samples $\bar{y}_i = G^{-1}(x_i)$ generated from source samples $x_i \sim \mu_0$. The inverse-map sampling path is detached, while $z(\bar{y}_i)$ is then evaluated through the native map G and differentiated normally. An approximation of the target μ for the forward KL and X_μ terms is supplied by the AIS construction in Section 3.2. When $\alpha > 0$, a second construction supplies samples of the wide-coverage measure $\hat{\mu}$ by quench and temper (Section 3.3). Throughout loss differentiation, the sampled coordinates and all resampling, mixture, and permutation choices are treated as fixed; gradients pass only through evaluations of z at the resulting coordinates. With equal target and mixture batch sizes, $\text{KL} + X_\mu + X_{(\hat{\mu} + \bar{\nu})/2}$ evaluates about twice as many samples per optimizer iteration as forward KL. This does not imply twice the total training time: constructing $\hat{\mu}$ also applies quench and temper to the validation set and adds a problem- and stage-dependent cost. Inference uses the same flow architecture for both methods.

Two recurring objectives are $\text{KL} + X_\mu$, with $\lambda = 1$ and $\alpha = \beta = 0$, and KLXX , with $\lambda = 1$ and $\alpha = \beta = \frac{1}{2}$. The first adds no sampling measure beyond the target batch, while the second draws the two components of its additional batch with equal probability. We report the full KLXX loss as $\text{KL} + X_\mu + X_{(\hat{\mu} + \bar{\nu})/2}$ when it must be distinguished from the other ablations.

We do not train with log-ratio variation alone, because empirical pairwise terms constrain relative log-ratios only on sampled regions and provide no global density-shaping constraint elsewhere. Appendix D.2 of the LDR study (Schopmans et al., 2026) shows the analogous failure: training with LDR-L1 or LDR-L2 alone, without a KL term, produces qualitatively incorrect distributions. Accordingly, every X-regularized objective in this paper includes forward KL.

3.2 Approximating μ by annealed importance sampling

Evaluating the μ -expectations in (16) requires samples from μ , which are not directly available. We form a surrogate by drawing pushforward samples $y_1, \dots, y_B$ from $\nu = (G^{-1})_{\#}\mu_0$ and reweighting them. In the theory, the normalized importance ratio is $w = \mu/\nu$. Computation uses the proportional weight

$$\tilde{w}(y) := \exp\left(U_0(G(y)) - U(y) - \log|\det J_G(y)|\right) \propto \frac{\mu(y)}{\nu(y)}, \quad y \in \mathbb{R}^d. \quad (17)$$

Below, w_i denotes these proportional weights because resampling, normalized ESS, and log-ratio differences are scale invariant. In high dimensions, however, μ and ν can differ greatly, so the weights can collapse onto a few outliers. We temper the reweighting through $M + 1$ bridge distributions

$$\pi_m(y) \propto \nu(y)^{1-\frac{m}{M}} \mu(y)^{\frac{m}{M}} \propto ((G^{-1})_{\#}\mu_0)(y) \tilde{w}(y)^{\frac{m}{M}}, \quad m = 0, 1, \dots, M, \quad (18)$$

so that $\pi_0 = \nu$ and $\pi_M = \mu$. Each level reweights by only $\tilde{w}(y)^{1/M}$, reducing the degeneracy of direct importance sampling. After each reweighting, we resample and diffuse the duplicates with the Metropolis-adjusted Langevin algorithm (MALA) (Roberts and Tweedie, 1996) before the next level.

Algorithm 1: Annealed importance sampling (AIS) for a biased target surrogate

Input: pushforward samples $y_1, \dots, y_B$ from $(G^{-1})_{\#}\mu_0$, target distribution μ , ladder length M

Output: biased target-surrogate batch $y_1, \dots, y_B$ approximating μ

```

1 for  $m = 1, \dots, M$  do
2   Compute the incremental weights  $\tilde{w}(y_i)^{1/M} = \exp(z(y_i)/M)$  on  $y_1, \dots, y_B$ ;
   // resample
3   Resample from the weighted set,  $y'_1, \dots, y'_B \leftarrow \text{resample}(y_{1:B}, (\tilde{w}(y_i)^{1/M})_{i=1}^B)$ ;
   // rejuvenate under the target
4   Diffuse the resampled duplicates by a short MALA chain on  $\mu$ ,  $y_{1:B} \leftarrow \text{MALA}(y'_{1:B}, U)$ ;
5 end
```

A standard SMC sampler targeting the path (18) would rejuvenate at level m with a Markov kernel invariant

for π_m . For $m < M$, a MALA kernel for π_m would require $\nabla_y \log \nu(y)$ and hence positional derivatives of the flow density, including $\nabla_y \log |\det J_G(y)|$; at $m = M$, this term disappears because $\pi_M = \mu$. Algorithm 1 instead applies MALA directly under μ , using only $-\nabla U$ in its proposal. MALA preserves μ , but not the intermediate π_m , so the output is a biased target surrogate. The construction trades intermediate invariance for a sampler that moves resampled samples under the final target. Section 2 studies a fixed smooth surrogate as a density-space idealization; it does not model the changing empirical output of Algorithm 1. Evaluation therefore uses fresh importance weights and coverage diagnostics.

The AIS construction in Algorithm 1 differs from FAB (Midgley et al., 2022). FAB places the flow inside the chains, and its MCMC transitions target the surrogate μ^2/ν , whose log-density gradient is

$$\nabla_y \log \frac{\mu^2(y)}{\nu(y)} = -2\nabla U(y) - \nabla_y \log \nu(y),$$

and therefore contains the positional gradient of the pushforward density at every MCMC step. FAB trains from a prioritized replay buffer of weighted AIS samples with a staleness correction. Reusing this buffer reduces repeated AIS generation but adds storage, prioritized sampling, and correction work. Here every MALA chain instead targets μ and uses only $-\nabla U$; the flow enters through the initial pushforward samples and evaluations of z . Each gradient step starts from a new source batch, so no replay buffer is kept.

The exploration costs also differ. Relative to forward KL, KLXX adds the QT construction and pairwise log-ratio variation evaluations: X_μ reuses the target batch, while the mixture term evaluates an additional batch. The fixed N -sample QT set of Section 3.3 supplies separate coverage information, whereas FAB relies on its AIS budget for exploration. FAB’s 2-divergence gives it a stronger tail-coverage incentive.

3.3 Constructing $\hat{\mu}$ by quench and temper

The last term of the loss (16) uses the wide-coverage measure $\hat{\mu}$ to improve candidate mode coverage when $\alpha > 0$. A finite source batch can omit remote target basins, and annealing initialized from that batch provides no log-ratio comparisons there unless its trajectories reach them.

QT addresses this limitation through three steps. The *melt* scatters the input samples across a wider region, the *quench* drives them toward mode centers with the limited-memory Broyden–Fletcher–Goldfarb–Shanno (L-BFGS) method (Liu and Nocedal, 1989), and the *temper* spreads them under the target around the discovered modes. Together, these steps construct a wide-coverage set that can include remote basins; $X_{\hat{\mu}}$ then adds comparisons on basins absent from the annealed batch, strengthening mode discovery beyond annealing alone. Formally, QT is the map

$$\hat{\mathcal{Y}} = \text{QT}(\mathcal{Y}, U)$$

from the validation set $\mathcal{Y} = \{x_1, \dots, x_N\}$ and target potential U to the N QT samples $\hat{\mathcal{Y}} = \{\hat{x}_1, \dots, \hat{x}_N\}$. Algorithm 2 applies this map once per stage attempt. The output $\hat{\mathcal{Y}}$ is held fixed as a reservoir during training. At each training step a mini-batch of B samples is drawn uniformly from that reservoir and refreshed by a short MALA chain before entering the X_ω computation. We use $\hat{\mu}$ for the law of this draw-and-refresh operation; its input is the uniform empirical measure on the fixed QT reservoir.

Algorithm 2: Quench and temper (QT): wide-coverage construction

Input: validation set $\mathcal{Y} = \{x_1, \dots, x_N\}$, target potential U , melt scale m_e

Output: wide-coverage set $\hat{\mathcal{Y}} = \{\hat{x}_1, \dots, \hat{x}_N\}$ forming $\hat{\mu}$, written $\hat{\mathcal{Y}} = \text{QT}(\mathcal{Y}, U)$

// melt

1 Scatter the samples $x_i \leftarrow x_i + m_e \zeta_i$ with $\zeta_i \sim \mathcal{N}(0, I_d)$ i.i.d. and melt scale $m_e > 0$;

// quench

2 Drive each sample toward a mode center of μ by L-BFGS on U , updating $x_1, \dots, x_N$;

// temper

3 Spread the samples around each mode, $\hat{x}_{1:N} \leftarrow \text{MALA}(x_{1:N}, U)$;

On a torus or Euclidean-torus product, periodic coordinates are interpreted modulo their period after the melt and after each MALA proposal. If the QT set contains a basin absent from the annealed target batch, $X_{\hat{\mu}}$ supplies pairwise log-ratio evaluations on that basin. QT does not certify complete mode discovery. A large melt scale m_e can bias the QT set toward basins far from the input set. This is intentional: $\hat{\mu}$ is a coverage measure, not an estimate of target mode probabilities. Adding $X_{\hat{\mu}}$ to the forward KL preserves $\nu = \mu$ as a minimizer, while empirical frequencies in the finite QT set affect the optimization path.

3.4 Evaluation: ESS, coverage, and importance sampling

We first assess the quality of the pushforward samples through two complementary diagnostics. The ESS measures density agreement through the observed importance weights on support reached by the flow. Coverage instead checks whether the pushforward samples reach the regions represented by a fixed QT reference set. A high ESS can coexist with a missing mode, while high coverage does not establish the correct relative density within the reached regions. We therefore report the two diagnostics together.

At evaluation time the trained flow G supplies the pushforward distribution $\nu = (G^{-1})_{\#}\mu_0$ and its tractable density $\nu(y) = \mu_0(G(y))|\det J_G(y)|$. The proportional importance weight $\tilde{w}(y)$ in (17) can therefore be evaluated on pushforward samples without fitting another density model. Draw $\mathbf{y} = (y_1, \dots, y_N)$ with $y_i = G^{-1}(x_i)$ and $x_i \sim \mu_0$ independently, and write $\mathbf{w} = (w_1, \dots, w_N)$ for any common scaling of the corresponding positive weights.

The first diagnostic is the ESS, normalized to $[0, 1]$ by dividing the classical ESS by the sample count N . For the weight vector $\mathbf{w}$, define

$$\text{ESS}(\mathbf{w}) = \frac{(\sum_{i=1}^N w_i)^2}{N \sum_{i=1}^N w_i^2} \in [0, 1]. \quad (19)$$

The upper bound follows from Cauchy–Schwarz and is attained when all weights are equal. The factor N distinguishes this normalized ESS from the classical count $(\sum_i w_i)^2 / \sum_i w_i^2 \in [1, N]$ (Elvira et al., 2022). A high ESS indicates uniform observed weights only on support reached by ν ; a target mode absent from the sample cannot lower it.

To detect such missing modes, we pair the ESS with the coverage diagnostic of Naeem et al. (2020). Let $\hat{\mathbf{y}} = (\hat{y}_1, \dots, \hat{y}_N) \in \mathbb{R}^{N \times d}$ be the fixed QT reference set produced by Algorithm 2. For each $1 \leq i \leq N$, the κ -nearest-neighbor radius of $\hat{y}_i$ (with $\kappa = 5$) is

$$\text{NND}_{\kappa}(\hat{y}_i; \hat{\mathbf{y}}) = \inf \left\{ r \geq 0 : \#\{1 \leq j \leq N : 0 < |\hat{y}_i - \hat{y}_j| < r\} \geq \kappa \right\}, \quad (20)$$

i.e., the distance from $\hat{y}_i$ to its κ -th nearest neighbor in $\hat{\mathbf{y}}$. The coverage of $\hat{\mathbf{y}}$ by the pushforward samples $\mathbf{y}$ is then

$$\text{Coverage}_{\kappa}(\mathbf{y}; \hat{\mathbf{y}}) = \frac{1}{N} \sum_{i=1}^N \mathbf{1} \left\{ \exists 1 \leq j \leq N : |y_j - \hat{y}_i| \leq \text{NND}_{\kappa}(\hat{y}_i; \hat{\mathbf{y}}) \right\}, \quad (21)$$

i.e., the fraction of points in $\hat{\mathbf{y}}$ whose κ -nearest-neighbor ball contains any of the pushforward samples. A coverage close to 1 indicates that the finite QT reference set is well covered by the flow, while a coverage well below 1 exposes missing regions that the ESS alone cannot see. Because the reference set is itself produced by QT, the diagnostic checks only modes found by that reference construction; it cannot reveal a mode absent from the reference set.

After these diagnostics assess the proposal, the same weights used for ESS define the resampling step. Normalize them as $p_i = w_i / \sum_{j=1}^N w_j$, resample N times from the pushforward samples with probabilities p_i , and apply a short MALA chain under μ to rejuvenate the resulting duplicates. The resampled samples form an unweighted approximation of the target-weighted empirical measure, while MALA produces fresh configurations and retains μ as its invariant distribution.

Under the support condition $\mu \ll \nu$ and standard consistency conditions, the target-weighted empirical measure and its resampled particle approximation converge to μ as $N \rightarrow \infty$ (Liu, 2001; Doucet et al., 2001). The μ -invariant MALA refresh preserves this target limit. If the pushforward samples contain no sample from a mode, however, resampling cannot create one and a short local refresh may not reach it. Coverage therefore remains necessary alongside ESS.

4 Adaptive-staging Boltzmann generator

A single flow can be inadequate when the source-to-target change is too large, so we divide the interpolation into stages. Each stage changes the potential by a controlled amount and trains one flow with the loss in (16), using user-chosen parameters $\lambda, \alpha, \beta \geq 0$. The KLXX Boltzmann generator is the special case $(\lambda, \alpha, \beta) = (1, \frac{1}{2}, \frac{1}{2})$. Sampling diagnostics determine the stage schedule.

4.1 Why staging must be adaptive

Two obstacles prevent a single flow from carrying μ_0 to μ in one map.

- *Curse of dimensionality.* In low dimension a unimodal source can often be reshaped into a multimodal target by one map, but as d grows the target landscape develops more basins, separated by higher barriers and occupying a smaller fraction of the source support. A normalizing flow of fixed depth and width can represent only a bounded change of shape, so the deformation it realizes in one training run is limited by its capacity rather than by the training budget. Requesting the whole source-to-target change from a single flow therefore leaves part of the target unfitted in exactly the high-dimensional, multimodal problems of interest. Splitting the interpolation into stages keeps each requested deformation within the capacity of one flow, and the number of stages needed to do so grows with dimension.
- *Structural limitation of a diffeomorphism.* A trained flow is a diffeomorphism of $\mathbb{R}^d$, so the pushforward ν inherits the connected support of μ_0 : the map deforms that support continuously and cannot split it into disconnected components. Mass therefore remains on the paths joining the target basins even when the basins themselves are well fitted, an effect visible between the isolated modes of the Sparse target in Figure 4. Reweighting and MALA rejuvenation at the interpolating distributions remove the leaked samples at every stage, so the support changes shape gradually across the schedule.

The stage schedule cannot be fixed in advance, because we assume no external information about the target. Training uses no target dataset, no reference trajectory, and no prior knowledge of the number, location, or depth of the potential wells. The only available information is oracle access to the target potential $U(x)$ and its gradient $\nabla U(x)$: the interpolating potentials are formed from U_0 and U alone, the loss evaluates U_t through the log-ratio z , the annealing and rejuvenation kernels use $-\nabla U_t$ in their proposals, and the quench of Algorithm 2 descends on the same potential. Under this restriction the difficulty of a candidate stage cannot be read off the potential before attempting it. The algorithm therefore probes the interpolation parameter: it proposes a stage increment, trains the stage flow, accepts the increment only when the resulting proposal passes a sampling gate on the validation set, and otherwise shrinks the increment and retries. Adaptive staging is thus more than a convenience. It is the mechanism that turns oracle access alone into a stage schedule matched to the difficulty the target actually presents, and it lets easy regions advance in few large stages while hard regions receive as many small stages as they require.

4.2 Construction of the adaptive stage schedule

To interpolate from the source $\mu_0 \propto \exp(-U_0)$ to the target $\mu \propto \exp(-U)$, we seek an increasing sequence

$$0 = t_0 < t_1 < \dots < t_K = 1$$

defining the interpolating potential and measure

$$U_t(x) = (1-t)U_0(x) + tU(x), \quad \mu_t(x) \propto \exp(-U_t(x)), \quad t \in [0, 1].$$

Here t is a dimensionless interpolation parameter, not a thermodynamic temperature. The discrete values t_k are stage points, and stage k is the transition from t_{k-1} to t_k . At the accepted stage points, write $\mu_k := \mu_{t_k}$, so μ_0 is the source and μ_K is the target. Stage k trains a target-to-source map G_k , whose inverse induces the proposal $\nu_k = (G_k^{-1})_{\#} \mu_{k-1}$ for μ_k . The number of stages K is selected during the run. The central scheduling choice is the stage increment $\Delta t_k = t_k - t_{k-1}$: a large increment can be hard to fit, whereas a small increment adds stages.

Training G_k requires the target surrogate for μ_k defined in Section 3.2. Before training, an optional classical SMC calculation can screen the proposed stage increment Δt_k . Its smallest incremental ESS measures overlap along that candidate stage for the current sample set. This is a difficulty diagnostic, not a mode-coverage guarantee. The screen floor τ_{SMC} lies in $[0, 1)$, and setting it to zero disables the screen. Throughout this section, the validation set is the sample set carried across stages and used in full for training batches, QT, and stage selection; it is not a held-out test set, and neither QT nor the SMC screen first draws a separately sized subset of it.

Algorithm 3: Adaptive selection of the stage increment Δt_k

Input: potentials U_0, U , stage point t_{k-1} , validation set $\mathcal{Y}_{k-1} = \{y_1, \dots, y_N\}$ approximating μ_{k-1} , ladder length M , SMC ESS floor $\tau_{\text{SMC}} = 0$, shrink factor $\gamma = 0.7$, initial guess t_{init}

Output: next stage point t_k , giving the stage increment $\Delta t_k = t_k - t_{k-1}$

```

1  $U_{t_{k-1}} \leftarrow (1 - t_{k-1})U_0 + t_{k-1}U;$ 
   // start from the supplied initial guess
2  $t_k \leftarrow t_{\text{init}};$ 
3 if  $\tau_{\text{SMC}} = 0$  then
4   | return  $t_k;$ 
5 end
6 while true do
   | // run a classical SMC ladder across the candidate stage
7   | Run an  $M$ -level SMC ladder on the set  $\mathcal{Y}_{k-1}$ , using the linear potential bridge from  $U_{t_{k-1}}$  to  $U_{t_k}$ ;
   | // evaluate the ladder by its worst per-level ESS
8   | Record  $\text{ESS}_{\min}$ , the smallest per-level ESS over the  $M$  ladder levels;
9   | if  $\text{ESS}_{\min} \geq \tau_{\text{SMC}}$  then
10  | | return  $t_k;$ 
11  | else
   | | // shrink the stage increment toward  $t_{k-1}$ 
12  | |  $t_k \leftarrow t_{k-1} + \gamma(t_k - t_{k-1});$ 
13  | end
14 end
```

When enabled, these SMC ladders involve only the potentials U_0 and U . They use no QT samples, so their per-level ESS can overlook a mode the SMC samples never reach. We therefore use this ESS only to screen a candidate stage; it is a heuristic measure of stage difficulty, not a coverage certificate. The selected-proposal validation in step (v) of Section 4 below supplies the acceptance gate, while QT supplies the separate wide-coverage set when $\alpha > 0$. By default $\tau_{\text{SMC}} = 0$, so Algorithm 3 returns t_{init} without running SMC. For a positive floor, it returns the first candidate stage along the shrink sequence whose estimated ESS clears that floor. The shrink factor is $\gamma = 0.7$. As t_k approaches t_{k-1} , the stage becomes trivial and its ESS approaches 1, so a floor below 1 is attainable under repeated shrinking in exact arithmetic.

4.3 The full training procedure

Algorithm 4 combines the stage-selection rule with the training procedure. It carries a fixed-size validation set $\mathcal{Y}_k \approx \mu_k$ across stages. Each candidate stage builds its QT set when $\alpha > 0$, trains a flow with the selected loss weights, and compares that flow with the identity proposal. If neither proposal reaches the validation ESS floor $\tau_v \in [0, 1)$, the algorithm shrinks the stage increment and retries. The displayed loop ends when $t_K = 1$. Practical implementations use an endpoint tolerance and finite caps on retries and stages; reaching a cap returns an incomplete stage schedule.

- (i) *Validation set.* Stage k uses the *validation set size* N and the *batch size* B , taken at least 2 so that the permuted-pair estimator has a pair to form. The validation set $\mathcal{Y}_{k-1} = \{y_1, \dots, y_N\} \approx \mu_{k-1}$ is carried over from the previous stage. Training batches are drawn at run time, B samples per gradient step, from $\mathcal{Y}_{k-1}$; no separate training set is built.
- (ii) *Adaptive stage selection.* Algorithm 3 selects the next stage point t_k . The first stage starts from $t_{\text{init}} = t_{\text{safe}} = 0.2$. Later stages extrapolate the last accepted stage increment, $t_{\text{init}} = \min(1, t_{k-1} + \Gamma(t_{k-1} - t_{k-2}))$, with enlarge factor $\Gamma = 1.5$. When a positive floor is supplied, the algorithm shrinks the stage increment until the classical SMC ESS on all N samples clears that floor. The selected stage target is $\mu_k \propto \exp(-U_{t_k})$, with $U_{t_k} = (1 - t_k)U_0 + t_k U$.
- (iii) *Wide-coverage set.* When $\alpha > 0$, quench and temper (Algorithm 2) acts on the validation set $\mathcal{Y}_{k-1}$ under U_{t_k} . The resulting N samples $\hat{\mathcal{Y}}_k = \text{QT}(\mathcal{Y}_{k-1}, U_{t_k})$ form the fixed stage reservoir. The measure $\hat{\mu}_k$ is the law obtained by drawing uniformly from this reservoir and applying the short MALA refresh used at each gradient step; it supplies the wide-coverage component of the last term in (22). When $\alpha = 0$, that component vanishes and the QT construction is omitted.
- (iv) *Training loss.* Each attempt initializes G_k to the identity map and trains it by the general loss (16) specialized to the stage $\mu_{k-1} \rightarrow \mu_k$. With the pushforward distribution $\nu_k = (G_k^{-1})_{\#} \mu_{k-1}$ and its frozen sampling measure $\bar{\nu}_k$, the stage log-ratio is

$$z_k(y) = U_{t_{k-1}}(G_k(y)) - U_{t_k}(y) - \log |\det J_{G_k}(y)|.$$

Let $s = \alpha + \beta$. For $s > 0$, define the stage specialization of the normalized mixture ξ , with any zero-weight component omitted,

$$\xi_k := \frac{\alpha \hat{\mu}_k + \beta \bar{\nu}_k}{s}.$$

The stage loss then reads

$$\mathcal{L}[G_k] = \underbrace{\mathbb{E}_{y \sim \mu_k} [z_k(y)]}_{\text{KL}(\mu_k \| \nu_k)} + \lambda \underbrace{\mathbb{E}_{y, y' \sim \mu_k} [z_k(y) - z_k(y')]}_{X_{\mu_k}(\mu_k \| \nu_k)} + s^2 \underbrace{\mathbb{E}_{y, y' \sim \xi_k} [|z_k(y) - z_k(y')|]}_{X_{\alpha \hat{\mu}_k + \beta \bar{\nu}_k}(\mu_k \| \nu_k)}. \quad (22)$$

When $s = 0$, the last term is omitted. The μ_k -expectations are estimated from the biased annealed surrogate: a batch $\mathcal{X}$ drawn from $\mathcal{Y}_{k-1}$ is mapped to the batch of pushforward samples $\tilde{\mathcal{X}}_{\nu} = G_k^{-1}(\mathcal{X})$, approximating ν_k , and annealed toward μ_k by Algorithm 1 with $G = G_k$. When $s > 0$, a batch $\mathcal{X}_{\xi_k}$ from ξ_k is formed by drawing from $\hat{\mu}_k$ with probability α/s when $\alpha > 0$, and from $\bar{\nu}_k$ with probability β/s when $\beta > 0$; the permuted-pair estimate is then multiplied by s^2 . This factor retains the general-weight definition in (4) when $\alpha + \beta \neq 1$. The choices $(\lambda, \alpha, \beta) = (0, 0, 0)$ and $(1, 0, 0)$ give forward KL and KL+ X_{μ} , respectively; $(1, \frac{1}{2}, \frac{1}{2})$ gives the default KLXX loss.

- (v) *Validation set update.* Once the candidate G_k is trained, each $y \in \mathcal{Y}_{k-1}$ is pushed through its inverse to $\tilde{y} = G_k^{-1}(y)$. This gives the trained set of pushforward samples $\tilde{\mathcal{Y}}_k^{\text{tr}} = G_k^{-1}(\mathcal{Y}_{k-1})$ from $\nu_k = (G_k^{-1})_{\#} \mu_{k-1}$, together with proportional importance weights represented by

$$w_k^{\text{tr}}(\tilde{y}) := \exp(z_k(\tilde{y})), \quad \tilde{y} \in \tilde{\mathcal{Y}}_k^{\text{tr}}.$$

The identity proposal keeps $\mathcal{Y}_{k-1}$ unchanged and has weights $w_k^{\text{id}}(y) = \exp[U_{t_{k-1}}(y) - U_{t_k}(y)]$. We select the pair with the larger ESS over the validation set and denote it by $(\tilde{\mathcal{Y}}_k, w_k)$, taking the stage map to be G_k or the identity accordingly. If $\text{ESS}(w_k) < \tau_v$, with default $\tau_v = 0.6$, we discard the candidate, shrink to $t_k \leftarrow t_{k-1} + \gamma(t_k - t_{k-1})$, and repeat steps (iii)–(v) at the reduced stage increment. Otherwise we advance the validation set to $\mathcal{Y}_k = \text{MALA}(\text{resample}(\tilde{\mathcal{Y}}_k, w_k; N), U_{t_k}) \approx \mu_k$.

The accepted stage schedule defines a Boltzmann generator for μ . The process starts with samples from μ_0 . At stage k , it applies the selected inverse map or identity, computes w_k , resamples, and rejuvenates under U_{t_k} . The resulting sample set supplies stage $k + 1$.

Algorithm 4: Adaptive-staging Boltzmann generator training

Input: potentials U_0, U ; sizes $N, B \geq 2$; loss weights $\lambda, \alpha, \beta \geq 0$; ladder length M ; ESS floors $\tau_{\text{SMC}} = 0$, $\tau_v = 0.6$; factors $\gamma = 0.7$, $\Gamma = 1.5$; safe start $t_{\text{safe}} = 0.2$

Output: completed stage schedule $0 = t_0 < \dots < t_K = 1$ and selected stage maps $\{G_k\}_{k=1}^K$

// validation set, $\mathcal{Y}_k \approx \mu_k$

```
1  $\mathcal{Y}_0 \leftarrow N$  samples from  $\mu_0$ ,  $t_0 \leftarrow 0$ ,  $k \leftarrow 0$ ;  
2 while  $t_k < 1$  do  
3    $k \leftarrow k + 1$ ;  
4   Set  $U_{t_{k-1}} \leftarrow (1 - t_{k-1})U_0 + t_{k-1}U$ ;  
   // (i) validation set  
5   Use  $\mathcal{Y}_{k-1}$  for stage selection, validation, and QT when  $\alpha > 0$ ;  
   // (ii) adaptive stage selection  
6   if  $k = 1$  then  
7     Set  $t_{\text{init}} = t_{\text{safe}}$ ;  
8   else  
9     Set  $t_{\text{init}} = \min(1, t_{k-1} + \Gamma(t_{k-1} - t_{k-2}))$ ;  
10  end  
11  Choose  $t_k$  by Algorithm 3 on  $\mathcal{Y}_{k-1}$  from  $t_{\text{init}}$ ;  
12  Set  $U_{t_k} \leftarrow (1 - t_k)U_0 + t_kU$ ;  
13  repeat  
14    Initialize  $G_k$  to the identity map;  
    // (iii) optional wide-coverage set  
15    If  $\alpha > 0$ , set  $\hat{\mathcal{Y}}_k \leftarrow \text{QT}(\mathcal{Y}_{k-1}, U_{t_k})$  by Algorithm 2;  
    // (iv) general log-ratio-variation loss  
16    for each gradient step do  
17      Draw  $B$  samples  $\mathcal{X}$  from  $\mathcal{Y}_{k-1}$ ;  
18      Set the batch of pushforward samples  $\bar{\mathcal{X}}_\nu \leftarrow G_k^{-1}(\mathcal{X})$  and detach its sampling path;  
19      Anneal these pushforward samples toward  $\mu_k$  to form the biased target batch  $\mathcal{X}_\mu$  (Algorithm 1);  
20      If  $\alpha > 0$ , draw  $\hat{\mathcal{X}}_\mu \subset \hat{\mathcal{Y}}_k$  and refresh it by MALA under  $U_{t_k}$ ;  
21       $s \leftarrow \alpha + \beta$ ; if  $s > 0$ , mix  $\hat{\mathcal{X}}_\mu$  (when  $\alpha > 0$ ) and  $\bar{\mathcal{X}}_\nu$  with probabilities  $\alpha/s, \beta/s$  to form  $\mathcal{X}_{\xi_k}$ ;  
22      Update  $G_k$  with the loss below using random batch permutations; omit its last term if  $s = 0$ :  
  
      
$$\hat{\mathcal{L}}[G_k] = \widehat{\mathbb{E}}_{y \in \mathcal{X}_\mu} [z_k(y)] + \lambda \widehat{\mathbb{E}}_{y, y' \in \mathcal{X}_\mu} [|z_k(y) - z_k(y')|] + s^2 \widehat{\mathbb{E}}_{y, y' \in \mathcal{X}_{\xi_k}} [|z_k(y) - z_k(y')|].$$
  
23    end  
    // (v) validation set update  
24    Form the trained and identity pairs as in step (v);  
25    Keep the larger-ESS pair as  $(\tilde{\mathcal{Y}}_k, w_k)$  and set  $G_k$  to the corresponding map;  
26    if  $\text{ESS}(w_k) < \tau_v$  then  
27      // reject the stage and shrink toward  $t_{k-1}$   
       $t_k \leftarrow t_{k-1} + \gamma(t_k - t_{k-1})$ , then update  $U_{t_k} \leftarrow (1 - t_k)U_0 + t_kU$ ;  
28    else  
29       $\mathcal{Y}_k \leftarrow \text{MALA}(\text{resample}(\tilde{\mathcal{Y}}_k, w_k; N), U_{t_k})$ ;  
30    end  
31  until  $\text{ESS}(w_k) \geq \tau_v$ ;  
32 end  
33  $K \leftarrow k$ ;  
34 return  $\{t_k\}_{k=0}^K$  and  $\{G_k\}_{k=1}^K$ ;
```

The accepted-stage weights also define the propagation factor used below:

$$F := \prod_{k=1}^K \text{ESS}(w_k)^{-1}. \quad (23)$$

Because normalized ESS lies in $(0, 1]$, F is at least one, with the ideal value one. It summarizes stagewise weight degeneracy by multiplying reciprocal incremental ESS values. It is neither the ESS of the complete composed sampler nor an estimator of endpoint Monte Carlo error. If a stage contains another accepted reweighting operation, such as molecular sharpening in Section 6, its reciprocal ESS enters the product.

5 Numerical experiments on model problems

We test forward KL against its X-augmented losses $\text{KL} + X_\mu$, $\text{KL} + X_\mu + X_{\hat{\mu}}$, and $\text{KL} + X_\mu + X_{(\hat{\mu} + \bar{\nu})/2}$, the last of which is KLXX. The target distributions vary in dimension, in the kind of barrier that separates their modes, and in how far those modes lie from the source. The 2D benchmarks separate finite-batch accuracy from mode discovery and make proposal leakage directly visible. The product multi-well raises the dimension to 256, where the multimodality is a product of independent sign choices. The tilted ϕ^4 model replaces isolated wells by a collective domain-wall barrier between two unequally weighted vacua. The clock model is a periodic 64-dimensional target with six symmetry-related sectors, trained by the full adaptive-staging construction. Accordingly, ESS is always reported together with a mode-coverage diagnostic, since neither quantity is conclusive on its own.

5.1 2D benchmark distributions

We consider four 2D targets: Two-Moon, Three-Well, Himmelblau, and Sparse. Two-Moon has two distorted crescent components and isolates finite-batch accuracy because both components lie within reach of the source. The remaining targets isolate mode collapse: Three-Well contains a low-probability basin, Himmelblau has four separated wells, and Sparse places two tight modes far into the source tails.

For every target, we initialize the same neural spline flow (NSF) (Durkan et al., 2019) at the identity and compare four objectives:

$$\text{KL}, \quad \text{KL} + X_\mu, \quad \text{KL} + X_\mu + X_{\hat{\mu}}, \quad \text{KL} + X_\mu + X_{(\hat{\mu} + \bar{\nu})/2}.$$

The flow has 32 spline bins, 6 transforms, and conditioner widths (128, 128). Each training step constructs a target surrogate by one annealing update through the current flow. The rejuvenation kernel uses 100 MALA steps of size 10^{-3} for Two-Moon and 50 MALA steps of size 2×10^{-3} for the other targets. The fixed validation sets contain 5×10^4 points for Two-Moon and Himmelblau and 8×10^4 for Three-Well and Sparse. For both objectives containing the wide-coverage term $X_{\hat{\mu}}$, quench and temper acts on the fixed validation set, with melt scale 2 and an Armijo L-BFGS quench using initial step 0.5 and 200 iterations. The final ESS is computed from flow importance weights on that validation set. Coverage, with $\kappa = 5$ against an independently generated quench and temper reference set, measures the fraction of reference configurations represented by the pushforward samples through the nearest-neighbor test of Section 3.4. The reference sets contain 1000, 4000, 500, and 4000 points for Two-Moon, Three-Well, Himmelblau, and Sparse, respectively; their QT tempering uses 1000 MALA steps at the target-specific step size above. Because coverage tests occupancy of the reference balls, a collapsed flow can have ESS near one but cannot have full coverage. Each objective-target pair is one fixed-seed training realization, so the numerical differences below describe these matched runs rather than variation across independent trainings.

Two-Moon. The source is $\mathcal{N}(0, I_2)$ and both crescents lie within its reach, so all four losses attain coverage 1. With batch size 500, 200 Adam steps, and learning rate 10^{-3} , forward KL reaches ESS 0.8358. Adding X_μ raises it to 0.9109, while the equal-weight mixture is best at 0.9236 (Figure 1). The loss $\text{KL} + X_\mu + X_{\hat{\mu}}$ stays at 0.8348, level with forward KL, since QT supplies no mode that the annealed batch is missing. Because coverage is already complete, this comparison tests whether pairwise log-ratio control improves density calibration on the reached support.

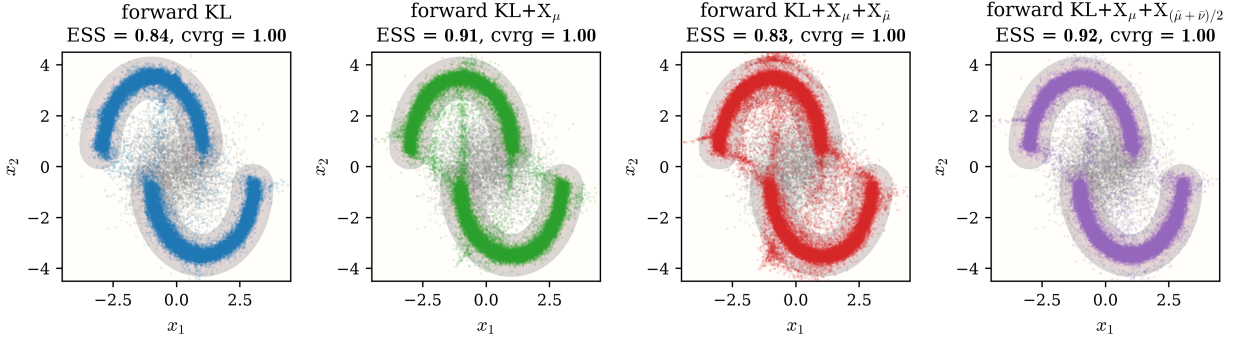


Figure 1: Two-Moon target. Columns, from left to right, are forward KL, $\text{KL}+X_\mu$, $\text{KL}+X_\mu+X_{\hat{\mu}}$, and $\text{KL}+X_\mu+X_{(\hat{\mu}+\bar{\nu})/2}$. Each panel overlays the pushforward samples on the target energy and reports final ESS and coverage. Gray points show the source.

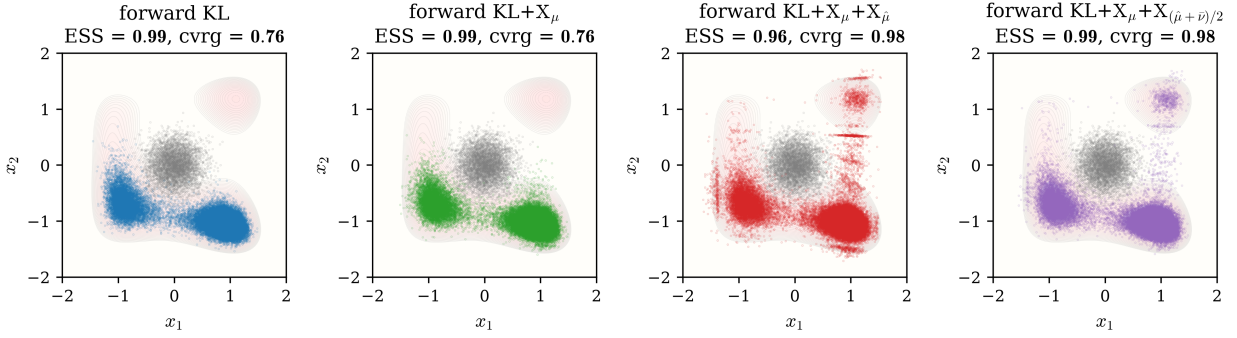


Figure 2: Three-Well target. From left to right, the panels show forward KL, $\text{KL}+X_\mu$, $\text{KL}+X_\mu+X_{\hat{\mu}}$, and $\text{KL}+X_\mu+X_{(\hat{\mu}+\bar{\nu})/2}$. Each panel overlays the pushforward samples on the target energy and reports final ESS and coverage; gray points show the source. Forward KL and $\text{KL}+X_\mu$ represent the two favored wells but miss the suppressed third well. Both objectives weighted by $\hat{\mu}$ place samples in all three wells, and the equal mixture retains the higher ESS of the two mode-complete fits.

Three-Well. The source is $\mathcal{N}(0, 0.25^2 I_2)$, narrower than the well separation and initially covering no basin. Training uses batch size 1000, 500 steps, and learning rate 10^{-3} . Forward KL and $\text{KL}+X_\mu$ fill the two favored wells and report ESS 0.9884 and 0.9902, respectively, while their coverage remains 0.7595. The QT set also represents the suppressed third well. Both objectives that use it cover all three wells in the sample panels, with scalar coverages 0.9783 and 0.9768 and ESS 0.9611 and 0.9893 (Figure 2).

Himmelblau. The source is $\mathcal{N}(0, I_2)$, whereas the four wells lie at radius approximately 3–4. Training uses batch size 1000, 1000 steps, and learning rate 10^{-3} . Forward KL and $\text{KL}+X_\mu$ capture only two wells, giving coverage 0.5160 despite ESS 0.9316 and 0.9809. Both objectives using $\hat{\mu}$ attain coverage 1. Relative to the $\text{KL}+X_\mu+X_{\hat{\mu}}$ result, the equal mixture shows less visible intermodal mass and raises ESS from 0.9562 to 0.9736 (Figure 3).

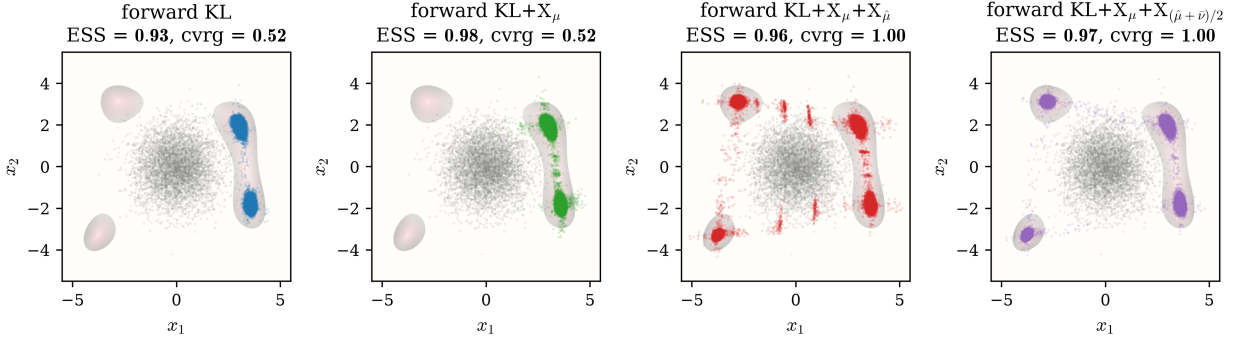


Figure 3: Himmelblau target. Columns show forward KL, $\text{KL}+X_\mu$, $\text{KL}+X_\mu+X_{\hat{\mu}}$, and $\text{KL}+X_\mu+X_{(\hat{\mu}+\bar{\nu})/2}$; each panel shows the pushforward samples over the target energy with final ESS and coverage, and gray points show the source. The first two objectives capture only two of the four wells. Both $\hat{\mu}$ -weighted objectives cover all four, while including pushforward samples in the mixture term reduces the visible intermodal mass and raises ESS relative to the $\text{KL}+X_\mu+X_{\hat{\mu}}$ fit.

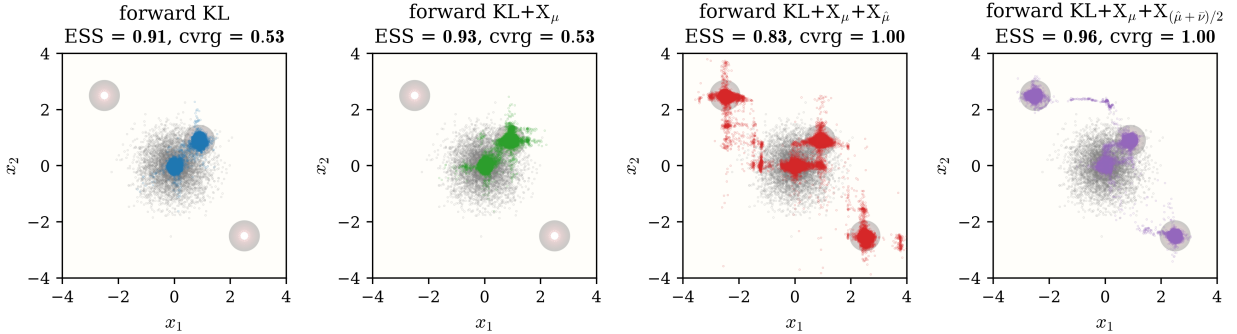


Figure 4: Sparse target. Columns show forward KL, $\text{KL}+X_\mu$, $\text{KL}+X_\mu+X_{\hat{\mu}}$, and $\text{KL}+X_\mu+X_{(\hat{\mu}+\bar{\nu})/2}$, with the pushforward samples overlaid on the target energy; panel titles report final ESS and coverage, and gray points show the source. The first two objectives cover the central pair but miss the two remote modes. The loss $\text{KL}+X_\mu+X_{\hat{\mu}}$ restores the remote modes with visible mass between components, whereas the equal mixture covers all four modes with less intermodal mass and higher ESS.

Sparse. The target is a four-component Gaussian mixture with component standard deviation 0.08: two modes lie near the source center and two lie in opposite corners, more than five source standard deviations away. With source standard deviation 0.6, batch size 1000, 1000 steps, and learning rate 5×10^{-3} , forward KL and $\text{KL}+X_\mu$ cover only the central pair. Their coverage is 0.5335, while their ESS values are 0.9116 and 0.9266. The loss $\text{KL}+X_\mu+X_{\hat{\mu}}$ restores all four modes but also places visible mass between them, with ESS 0.8262. The equal mixture attains coverage 1, shows less intermodal mass, and reaches ESS 0.9555.

The four targets support the intended division of labor. Adding X_μ raises ESS when coverage is already complete. QT samples supply information about modes absent from the annealed batch. Including pushforward samples in the mixture term is associated with cleaner intermodal regions than the loss $\text{KL}+X_\mu+X_{\hat{\mu}}$. The equal mixture has the highest mode-complete ESS on every target in Table 2.

5.2 High-dimensional product multi-well

We next isolate accuracy scaling over the dimensions $d = 16, 32, \dots, 256$. Letting $k = \lfloor \log_2 d \rfloor$, we use

$$U(\mathbf{x}) = \frac{1}{2} \|\mathbf{x}\|^2 + 12 \sum_{i=1}^k \exp(-x_i^2). \quad (24)$$

target	KL	KL+ X_μ	KL+ $X_\mu+X_{\hat{\mu}}$	KL+ $X_\mu+X_{(\hat{\mu}+\hat{\nu})/2}$
Two-Moon	0.836 (1.000)	0.911 (1.000)	0.835 (1.000)	0.924 (1.000)
Three-Well	0.988 (0.760)	0.990 (0.760)	0.961 (0.978)	0.989 (0.977)
Himmelblau	0.932 (0.516)	0.981 (0.516)	0.956 (1.000)	0.974 (1.000)
Sparse	0.912 (0.534)	0.927 (0.534)	0.826 (1.000)	0.956 (1.000)

Table 2: Final ESS, with $\kappa = 5$ coverage in parentheses, on the four 2D targets. Bold marks the highest ESS among methods whose sample panel covers every target mode. The scalar coverage diagnostic slightly undercounts the visibly occupied third mode on Three-Well.

The first k coordinates are symmetric double wells with minima at $\pm\sqrt{\ln 24}$ and barrier approximately 9.9, while the remaining $d - k$ coordinates are standard Gaussian. The target therefore has exactly 2^k equal-weight sign-pattern modes, where $2^k \leq d < 2^{k+1}$. Multimodality occupies only a k -dimensional subspace, but the log-weight error accumulates across all d coordinates.

All four objectives use the same identity-initialized NSF on $[-4, 4]^d$, with 16 bins, 6 transforms, and conditioner widths (256, 256). At every dimension, training uses batch size 250, 2000 Adam steps at learning rate 10^{-3} , gradient clipping at 10^2 , and a fixed validation set of size $5000d$. The live target surrogate uses one annealing update followed by 50 MALA steps of size 2×10^{-3} . This is a standalone one-flow ablation, so its separately reported QT set contains $1000d$ samples rather than reusing the validation set as the staged algorithm does. QT uses melt scale 2 and an Armijo L-BFGS quench with initial step 0.5 and 200 iterations. Final validation ESS is evaluated on $\max(200d, 80000)$ fresh source samples. Each objective–dimension pair is one fixed-seed training realization.

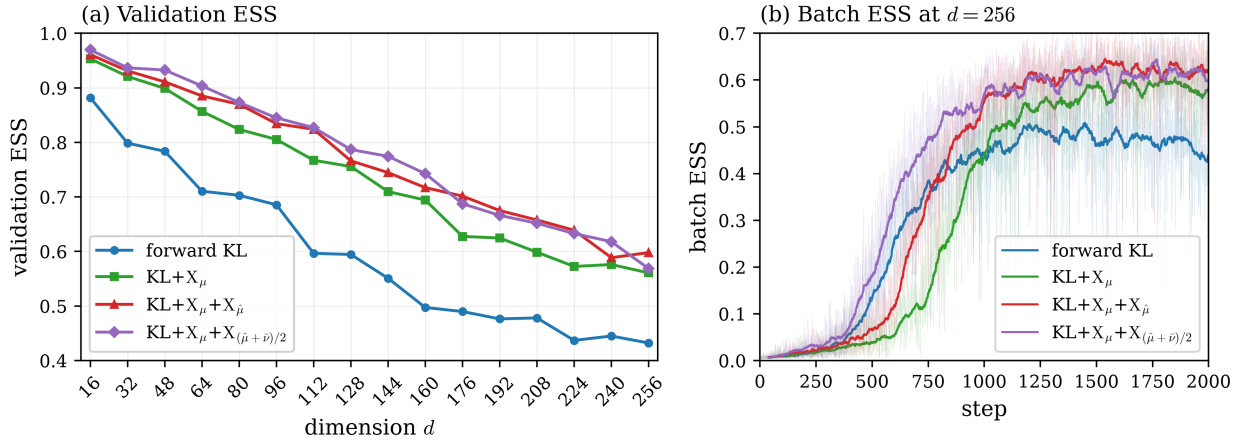


Figure 5: ESS on the product multi-well target (24). (a) Final validation ESS over all sixteen tested dimensions. (b) Batch ESS during the $d = 256$ training run; faint curves are raw values and bold curves are 40-step moving averages.

In each of the 64 evaluated pairs of objective and dimension, all 2^k sign patterns of the double-well coordinates occur among the fresh evaluation samples, including all 256 patterns for each of the four objectives at $d = 256$. The ESS comparison is therefore not explained by a missing sign-pattern mode in those samples. Figure 5 shows that each of the three objectives with log-ratio variation has higher final validation ESS than forward KL at every one of the sixteen dimensions, so forward KL is the lowest of the four at each dimension. At $d = 256$, the final values are 0.4322 for forward KL, 0.5608 for KL+ X_μ , 0.5979 for KL+ $X_\mu+X_{\hat{\mu}}$, and 0.5687 for KL+ $X_\mu+X_{(\hat{\mu}+\hat{\nu})/2}$. The two objectives containing $\hat{\mu}$ behave similarly across dimensions. During the $d = 256$ training realization, all three objectives with log-ratio variation continue to improve after forward KL begins to plateau.

5.3 Lattice ϕ^4 field theory: ESS at a collective barrier

The tilted scalar ϕ^4 theory tests ESS at a collective barrier. On a periodic $L \times L$ lattice, with $d = L^2$ and magnetization $m(\phi) = d^{-1} \sum_j \phi_j$, the action is

$$U(\phi) = -0.8 \sum_{\langle jl \rangle} \phi_j \phi_l + \sum_{j=1}^d \left[\phi_j^2 + \frac{1}{2}(\phi_j^2 - 1)^2 \right] + h \sum_{j=1}^d \phi_j.$$

We take $h = 0.0257$ at $L = 6$ and $h = 0.0144$ at $L = 8$. This size-dependent field keeps the extensive tilt scale nearly fixed: $hL^2 = 0.9252$ at $L = 6$ and $hL^2 = 0.9216$ at $L = 8$. Thus the two targets have comparable phase imbalance despite their different volumes. In the broken phase the field orders near two vacua $\pm v\mathbf{1}$, where $v > 0$ is the spontaneous field magnitude and $\mathbf{1} \in \mathbb{R}^d$ is the all-ones vector. A collective domain-wall barrier separates the vacua. The positive linear term gives the two phases unequal probability, so the minority weight $p_+ = \mathbb{P}(m > 0)$ is not fixed by symmetry and must be learned.

Parallel tempering gives the references $p_+ = 0.1251 \pm 0.0009$ and 0.1284 ± 0.0014 , statistically consistent with occupancies 0.1260 and 0.1268 from an independent mirror-MALA construction. That construction samples the positive vacuum of the symmetric $h = 0$ action, obtains the negative vacuum by the exact reflection $\phi \mapsto -\phi$, reweights the pair by the field term, and resamples one million configurations. Each loss is run at seeds 0, 1, 2 using an identity-initialized NSF on $[-3, 3]^d$, with 16 bins, 6 transforms, and conditioner widths (256, 256). The source is $\mathcal{N}(0, 0.25I_d)$, centered on the barrier. Training uses a fixed validation set of 10^5 source samples, batch size 500, 2000 Adam steps at learning rate 10^{-3} , and gradient clipping at 10^3 . The live surrogate uses one annealing update followed by 50 MALA steps of size 2×10^{-3} . For both objectives containing $X_{\hat{\mu}}$, quench and temper acts on the validation set, with melt scale 2 and an Armijo L-BFGS quench using initial step 0.1 and 200 iterations.

	$L = 6$		$L = 8$	
	ESS	p_+	ESS	p_+
forward KL	0.794/0.840/0.826	1/1/0	0.622/0.686/0.684	1/0/1
KL+ X_μ	0.935/0.933/0.945	1/1/0	0.804/0.824/0.821	1/0/0
KL+ X_μ + $X_{\hat{\mu}}$	0.875/0.873/0.871	0.127/0.128/0.127	0.636/0.645/0.654	0.125/0.124/0.124
KL+ X_μ + $X_{(\hat{\mu}+\bar{v})/2}$	0.896/0.877/0.894	0.127/0.128/0.127	0.690/0.676/0.681	0.131/0.130/0.129
PT reference	—	0.1251 ± 0.0009	—	0.1284 ± 0.0014

Table 3: Final ESS and reweighted minority-phase weight p_+ for three seeds on the two tilted ϕ^4 targets, $(L, h) = (6, 0.0257)$ and $(8, 0.0144)$. The size-dependent field keeps $hL^2 \simeq 0.923$, making the phase imbalance comparable across the two volumes. Integer values 0 and 1 indicate collapse onto one vacuum; bold marks the higher ESS among the objectives that recover both vacua.

Forward KL and KL+ X_μ collapse at every size and seed. Their annealed batches start from pushforward samples and do not cross the collective barrier within the run, so the second vacuum remains absent and reweighting gives $p_+ = 0$ or 1. Nevertheless, KL+ X_μ has the highest ESS at both sizes: it improves calibration on the reached vacuum while leaving the missing phase undetected by ESS.

Both objectives using QT populate both vacua at every seed. Their recovered p_+ lies between 0.1238 and 0.1312, within 0.005 of the parallel-tempering estimate and the resampled mirror-MALA reference at the corresponding lattice size. In these finite-capacity fits, recovering both vacua is accompanied by some mass in the suppressed region between phases and lower ESS. At $L = 8$, the equal mixture has ESS 0.6756–0.6902, compared with 0.6355–0.6541 for the loss KL+ X_μ + $X_{\hat{\mu}}$. Its samples retain both vacua with less visible mass in the suppressed region between phases, consistent with the intended role of the pushforward samples.

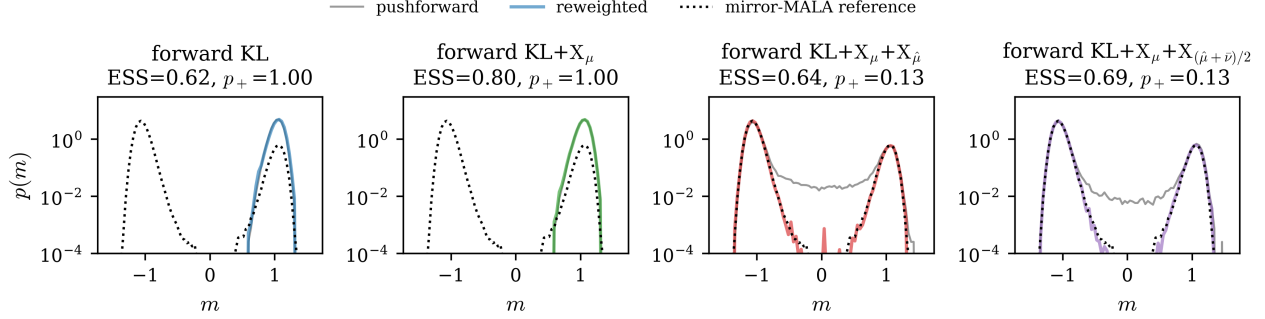


Figure 6: Magnetization density on the $L = 8$ lattice at seed 0. Each panel shows the pushforward density (gray), the reweighted density (color), and the resampled mirror-MALA reference density (dotted); the title reports ESS and the reweighted p_+ .

5.4 Boltzmann generator on p -state clock model

The preceding benchmarks use a single flow. We now test the full adaptive-staging construction on the $p = 6$ clock model on a periodic 8×8 lattice,

$$U(\boldsymbol{\theta}) = - \sum_{\langle jl \rangle} \cos(\theta_j - \theta_l) - \frac{1}{2} \sum_{j=1}^d \cos(6\theta_j), \quad \boldsymbol{\theta} \in [-\pi, \pi)^d, \quad d = 64, \quad (25)$$

where $\langle jl \rangle$ runs over nearest-neighbor pairs. The source is uniform on the torus and the interpolation is $U_t = tU$. At low temperature, the target separates into six symmetry-related clock sectors.

Each stage uses an identity-initialized neural circular spline flow with 16 bins, 6 autoregressive transforms, and conditioner widths (256, 256). The validation set contains 4×10^5 samples, and both SMC and quench and temper act directly on this set. Every stage uses 3000 Adam steps at learning rate 10^{-3} , a four-level annealed surrogate with 100 MALA steps of size 10^{-3} , energy clipping at 10^3 , and gradient clipping at 10^2 . For KLXX, QT uses wrapped-Gaussian melt scale 2π and an Armijo L-BFGS quench with initial step 10^{-2} and 100 iterations. The clock stage schedule uses the nondefault settings $t_{\text{safe}} = 0.25$, enlarge factor $\Gamma = 2$, and gates $\tau_{\text{SMC}} = 0.7$ and $\tau_v = 0.4$. We report batch sizes $B = 2000, 1000, 500, 250$. An additional completed run lies below this reporting cutoff and is excluded from the table and all aggregate clock claims.

The comparison is schedule-matched. An adaptive stage schedule is first constructed with forward KL using SMC preselection and an ESS acceptance gate. The loss $\text{KL} + X_\mu + X_{(\hat{\mu} + \bar{\nu})/2}$ is then trained on the identical accepted values of t , without a second gate. Hence the two methods see the same stage increments within each batch size pair. Every configuration is a single run, so individual stage ESS values carry single-run fluctuation.

The principal diagnostic is the selected-proposal validation ESS of each accepted stage. The propagation factor in (23) summarizes stagewise weight degeneracy across the stage sequence; smaller is better. It is not a full-chain ESS or an endpoint error estimate. Table 4 reports every accepted stage and the resulting factor.

The loss $\text{KL} + X_\mu + X_{(\hat{\mu} + \bar{\nu})/2}$ raises the geometric mean stage ESS at every reported batch size: from 0.556 to 0.665 at $B = 2000$, from 0.553 to 0.643 at $B = 1000$, from 0.487 to 0.571 at $B = 500$, and from 0.470 to 0.553 at $B = 250$. Correspondingly, it reduces the factor by ratios between 2.87 and 3.70. The gain is not confined to the small-batch regime: at the largest batch, the factor falls from 33.9 to 11.6.

The comparison is also consistent stage by stage. The loss $\text{KL} + X_\mu + X_{(\hat{\mu} + \bar{\nu})/2}$ has higher validation ESS on all 28 shared stages, including every stage in the large-batch runs. As the batch shrinks, the adaptive stage schedule constructed with forward KL lengthens from six to eight reported stages and both propagation factors grow. KLXX nevertheless accumulates less stagewise weight degeneracy before the final target is reached.

stage k	1	2	3	4	5	6	7	8	F
t_k ($B = 2000$)	0.250	0.495	0.663	0.779	0.887	1.000	—	—	
forward KL	0.751	0.570	0.471	0.474	0.511	0.602	—	—	33.9
KL+ $X_\mu+X_{(\hat{\mu}+\bar{\nu})/2}$	0.921	0.759	0.608	0.562	0.558	0.649	—	—	11.6
t_k ($B = 1000$)	0.250	0.495	0.613	0.728	0.841	0.952	1.000	—	
forward KL	0.672	0.479	0.580	0.463	0.427	0.520	0.830	—	63.0
KL+ $X_\mu+X_{(\hat{\mu}+\bar{\nu})/2}$	0.866	0.642	0.700	0.545	0.462	0.541	0.859	—	22.0
t_k ($B = 500$)	0.250	0.421	0.590	0.705	0.784	0.895	1.000	—	
forward KL	0.580	0.549	0.408	0.432	0.525	0.417	0.530	—	153.5
KL+ $X_\mu+X_{(\hat{\mu}+\bar{\nu})/2}$	0.778	0.695	0.506	0.500	0.595	0.429	0.566	—	50.7
t_k ($B = 250$)	0.250	0.421	0.539	0.654	0.734	0.811	0.887	1.000	
forward KL	0.468	0.448	0.495	0.428	0.506	0.484	0.514	0.424	421.2
KL+ $X_\mu+X_{(\hat{\mu}+\bar{\nu})/2}$	0.659	0.586	0.601	0.480	0.563	0.536	0.564	0.464	113.8

Table 4: Selected-proposal validation ESS on every accepted stage of the schedule-matched clock-model comparison between forward KL and KL+ $X_\mu+X_{(\hat{\mu}+\bar{\nu})/2}$. Within each batch size block, both losses traverse the listed stage schedule. The final column is the propagation factor in (23); bold marks the better loss at each stage and the smaller propagation factor.

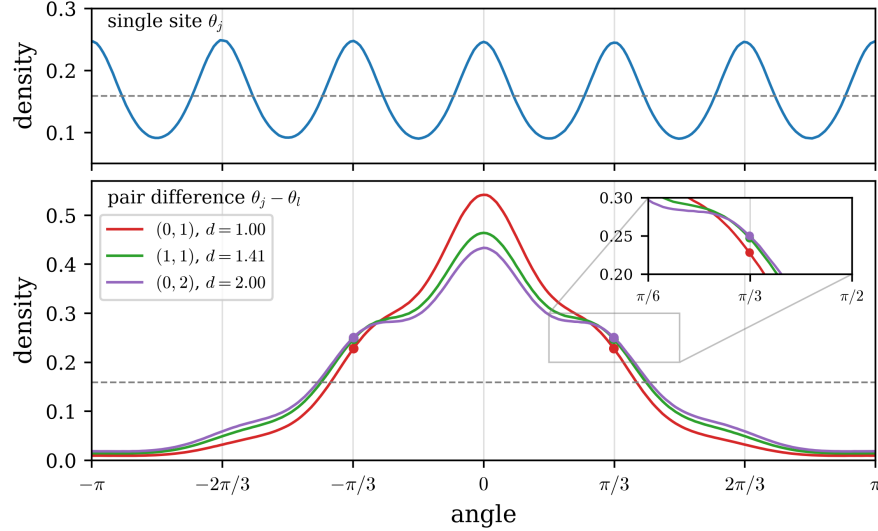


Figure 7: Angle densities from 2×10^6 fresh samples advanced through the trained $B = 2000$ KL+ $X_\mu+X_{(\hat{\mu}+\bar{\nu})/2}$ stage sequence by map, reweighting, resampling, and MALA. The upper panel is the single-site marginal. The lower panel gives pair differences at offsets (0, 1), (1, 1), and (0, 2); dots and the inset resolve the one-clock-step angle $\pi/3$. Dashed lines show the uniform density.

The single-site marginal exhibits all six clock wells with nearly equal peak heights. Pair differences are most concentrated at zero for nearest neighbors; as the lattice separation increases, the alignment peak falls and the shoulders near $\pm\pi/3$ gain mass. This is the expected decay of local angular order with distance and confirms that the staged sampler is not confined to a single global sector.

To quantify whether staged resampling preserves the sixfold symmetry, we perform two fresh-rebuild tests through the trained stage sequences. The batch size test fixes $N = 6.4 \times 10^5$, uses $B \in \{2000, 1000, 500, 250\}$, and averages over four matched sampling seeds for each method and batch size. The particle-count test fixes $B = 2000$ and uses $N = 10^4 2^k$, $k = 0, \dots, 6$, with 2^{8-k} independent rebuilds at each size so every particle-count value has equal total work of 2.56×10^6 particles. For each sample, let s be the clock sector nearest to the argument of the complex magnetization $d^{-1} \sum_{j=1}^d e^{i\theta_j}$, and let p_s be the empirical fraction assigned to sector s . We define

$$\text{occupancy bias} := \frac{1}{6} \sum_{s=0}^5 \left| p_s - \frac{1}{6} \right|.$$

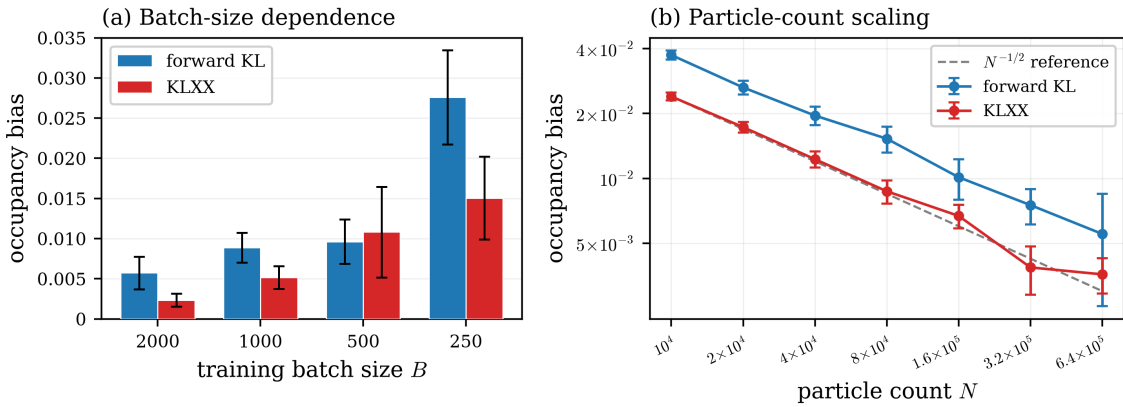


Figure 8: Clock-sector occupancy bias under two fresh-rebuild tests. (a) Mean bias at fixed $N = 6.4 \times 10^5$ across training batch sizes; each bar averages four matched sampling seeds. (b) Mean bias at fixed $B = 2000$ versus particle count; at $N = 2^k \times 10^4$, each mean uses 2^{8-k} independent rebuilds, keeping the total work per point at 2.56×10^6 particles. Error bars in both panels are ± 2 standard errors. The dashed line in (b) is an $N^{-1/2}$ reference anchored at the smallest $\text{KL} + X_\mu + X_{(\hat{\mu} + \hat{\nu})/2}$ point. The panels use separately generated rebuild ensembles.

At fixed N , panel (a) shows that the mean occupancy bias increases as the training batch shrinks for both methods. KLXX has the lower mean at $B = 2000, 1000$, and 250 . At those batch sizes, the forward KL means are 0.00573, 0.00888, and 0.02760, while the KLXX means are 0.00235, 0.00517, and 0.01506. At $B = 500$ the ordering reverses slightly, from 0.00963 for forward KL to 0.01082 for KLXX, and the intervals defined by ± 2 standard errors overlap.

At fixed $B = 2000$, panel (b) follows the Monte Carlo rate: the log-log slopes of occupancy bias against N are -0.459 for forward KL and -0.479 for $\text{KL} + X_\mu + X_{(\hat{\mu} + \hat{\nu})/2}$. The absence of a visible bias floor is consistent with sampling fluctuation rather than a particle-count-independent sector imbalance over the tested range. The KLXX curve is lower at all seven reported sizes, reducing the mean bias from 0.03735 to 0.02397 at $N = 10^4$ and from 0.00552 to 0.00358 at $N = 6.4 \times 10^5$. Every reported rebuild in both panels retains all six sectors. Together with the six peaks in Figure 7, these tests show that the staged samples are not confined to one global sector. The rebuild seeds measure sampling variability; each stage schedule for a given method and batch size is one training run, so the between-method differences do not estimate variation over independently trained models.

6 Boltzmann generators on molecular potential energies

The model problems above separate the target and QT terms on explicit targets and probe the term built from pushforward samples. Molecular systems add a singular nonbonded core, a mixed Euclidean-periodic domain, and more conformational degrees of freedom as molecular size grows. We first define the molecular measure and its regularized potential, then add sharpening to the staged generator and report the completed alkane family, three additional achiral molecules, and three chiral KLXX runs with fixed stereochemical configurations.

6.1 Molecular potential models

For Cartesian coordinates x , let $E(x)$ denote the complete molecular-mechanics energy

$$E(x) = E_{\text{bond}} + E_{\text{angle}} + E_{\text{torsion}} + E_{\text{LJ}} + E_{\text{Coul}} + E_{\text{GB}}^{\text{OBC1}} + E_{\text{ACE}}. \quad (26)$$

The bonded and vacuum nonbonded part has the Amber form (Cornell et al., 1995)

$$\begin{aligned} E_{\text{vac}}(x) = & \sum_b k_b (r_b - r_b^0)^2 + \sum_a k_a (\theta_a - \theta_a^0)^2 + \sum_\tau \frac{V_\tau}{2} [1 + \cos(n_\tau \varphi_\tau - \gamma_\tau)] \\ & + \sum_{\substack{i < j \\ \text{nonbonded}}} \left\{ 4\epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right] + k_e \frac{q_i q_j}{r_{ij}} \right\}, \end{aligned} \quad (27)$$

Here b , a , and τ index bonds, angles, and proper or improper torsions, while i, j index nonbonded atom pairs after topology-defined exclusions and scaling. The symbols r_b^0 , θ_a^0 , k_b , k_a , V_τ , n_τ , and γ_τ are force-field equilibrium values and coefficients; σ_{ij} , ϵ_{ij} , and q_i are the nonbonded parameters. OBC1 supplies the generalized-Born electrostatic solvation term (Onufriev et al., 2004), and ACE supplies the nonpolar term (Schaefer and Karplus, 1996). All terms in (26) belong to the sampled potential.

A nonlinear molecule with A atoms has $d = 3A - 6$ internal degrees of freedom after rigid translation and rotation are removed. A fixed internal-coordinate tree reconstructs $x = x(q)$ from whitened bond and angle coordinates and periodic torsions. Write d_E and d_T for the numbers of nonperiodic Euclidean coordinates and periodic torsions, respectively, and let $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$. Consequently,

$$\mathcal{Q} = \mathbb{R}^{d_E} \times \mathbb{T}^{d_T}, \quad d_E + d_T = 3A - 6, \quad (28)$$

and the physical reduced potential at inverse temperature $\beta_T = (k_B T)^{-1}$ is

$$U(q) = \beta_T E(x(q)) - \log J(q), \quad (29)$$

where $J(q)$ is the Cartesian volume Jacobian of the reconstruction, including the whitening maps. The source is Gaussian on the Euclidean block and uniform on the torus. The mixed spline flow uses ordinary rational-quadratic splines on the Euclidean coordinates and circular splines on the torsions. We abbreviate *N*-acetyl-L-proline *N*-methylester as Ac-Pro-NHMe.

The alkane family uses neutral all-atom bundles with GAFF2 parameters, AM1-BCC charges, graph-symmetry and neutrality projection, and mbondi2 radii. GAFF and AM1-BCC are described in Wang et al. (2004) and Jakalian et al. (2002); the bundles are prepared with AmberTools (Case et al., 2023). OBC1/ACE is used across the homologous series, and the independent references are propagated with OpenMM (Eastman et al., 2017). NMA uses the 12-atom ACE-NME ff96/OBC1/ACE Hamiltonian. Glycerol, neutral diethanolamine, and (2*R*, 3*R*)-2,3-butanediol use GAFF2/AM1-BCC with OBC1/ACE. Alanine dipeptide uses the 22-atom ACE-ALA-NME Hamiltonian of the FAB benchmark (Midgley et al., 2022): Amber ff96, OBC1 with igb = 2, mbondi2 radii, solute and solvent dielectric constants 1 and 78.5, and the L-alanine component selected by the reference trajectory. Ac-Pro-NHMe uses the corresponding 26-atom ACE-PRO-NME ff96/OBC1/ACE Hamiltonian.

The alkanes, glycerol, neutral diethanolamine, and NMA have no fixed configurational stereocenter under these bundle definitions. Neutral diethanolamine retains both signs of its nitrogen-pyramidal coordinate,

target	atoms / d	model
<i>Achiral targets</i>		
alkanes, C ₁ –C ₆	5–20/9–54	GAFF2/AM1-BCC; OBC1/ACE
<i>N</i> -methyleacetamide (NMA)	12/30	ff96; OBC1/ACE
glycerol	14/36	GAFF2/AM1-BCC; OBC1/ACE
neutral diethanolamine	18/48	GAFF2/AM1-BCC; OBC1/ACE
<i>Chiral targets: one fixed stereochemical configuration</i>		
alanine dipeptide	22/60	ff96; OBC1/ACE
(2 <i>R</i> , 3 <i>R</i>)-2,3-butanediol	16/42	GAFF2/AM1-BCC; OBC1/ACE
Ac-Pro-NHMe	26/72	ff96; OBC1/ACE

Table 5: Molecular targets in mixed internal coordinates, grouped by configurational stereochemistry. All systems are at 300 K, use no nonbonded cutoff or bond constraints, and include the coordinate Jacobian in (29). For each chiral target, the generator and reference are restricted to one specified stereochemical configuration, while all conformations within that configuration remain part of the target.

and the NMA amide torsion remains periodic so that both cis and trans conformations belong to its target. The chiral targets retain only the named configurations: L-alanine for alanine dipeptide, (2*R*, 3*R*) for 2,3-butanediol, and L-proline for Ac-Pro-NHMe. All conformational basins within each selected configuration remain part of the corresponding target. These support conventions are part of the target rather than an after-the-fact filter.

The Lennard–Jones core in (27) grows as r^{-12} , so early flow proposals can place a few samples at energies and gradients far beyond the scale of the thermally occupied region. We therefore use a two-parameter molecular surrogate. Let $x_\star$ be the bundle reference geometry. After flooring only the regular and exception Coulomb/Lennard–Jones pair distances at r , denote the complete force-field energy by $E_r(x)$, its reference value by $E_r^\star := E_r(x_\star)$, and its reference-relative value by $\Delta E_r(x) := E_r(x) - E_r^\star$. For an energy threshold e , define the soft cap

$$\mathcal{C}_e(\Delta E) = \begin{cases} \Delta E, & \Delta E \leq e, \\ e[1 + \log(\Delta E/e)], & \Delta E > e, \end{cases} \quad (30)$$

and collect the two regularization parameters in $\rho = (e, r)$. The corresponding reduced potential is

$$U^{\{\rho\}}(q) = \beta_T [E_r^\star + \mathcal{C}_e(\Delta E_r(x(q)))] - \log J(q). \quad (31)$$

The threshold e is measured in kJ mol^{−1}, and the pair floor r is measured in nm. The component formulas of the bonded and OBC1/ACE terms are unchanged before the scalar cap is applied to the complete energy. The coordinate Jacobian and domain are not capped. The physical potential U in (29) is recovered in the formal limit $e \rightarrow \infty$, $r \rightarrow 0$. Thus $U^{\{\rho\}}$ is a regularized sampling potential, not optimizer clipping or a replacement for the physical energy used in the observables.

6.2 Boltzmann generators with sharpening

A softer molecular potential can make early stages easier in a given adaptive-staging run. After each flow stage, *sharpening* reweights the validation set toward a stricter regularization setting, resamples it, and applies MALA under the stricter potential. This is a stochastic potential update, not an additional flow-training loss. To extend the source-to-target interpolation of Section 4, let $\rho_0 = (e_0, r_0)$ and $\rho_1 = (e_1, r_1)$ be the soft and final settings and interpolate them by

$$\rho_s = (1 - s)\rho_0 + s\rho_1, \quad s \in [0, 1]. \quad (32)$$

The alkane and NMA experiments use the finite endpoint $\rho_1 = (100, 0.15)$, while glycerol and neutral diethanolamine use $\rho_1 = (100, 0.10)$. The residual difference from U is audited separately. A fixed-regularization run has $\rho_0 = \rho_1$. Extending the interpolation of Section 4 by the single index s , set

$$U_{t,s}(q) = (1 - t)U_0(q) + tU^{\{\rho_s\}}(q), \quad \mu_{t,s} \propto \exp(-U_{t,s}). \quad (33)$$

Thus t remains the source-to-target interpolation parameter defined in Section 4, while s changes only the regularization setting. If s is fixed, suppressing the second index recovers the notation U_t, μ_t of Section 4.

Accepted sample sets lie on the diagonal $s = t$, but one stage advances the two coordinates in sequence:

$$\mathcal{Y}_{k-1} \approx \mu_{t_{k-1}, t_{k-1}} \xrightarrow[\text{selected } (\hat{\mathcal{Y}}_k, w_k)]{\text{flow or identity}} \mathcal{Y}_k^- \approx \mu_{t_k, t_{k-1}} \xrightarrow[w_k^{\text{sharp}}]{\text{sharpen}} \mathcal{Y}_k \approx \mu_{t_k, t_k}. \quad (34)$$

The first arrow holds $s = t_{k-1}$ fixed and advances t from t_{k-1} to t_k . Potential-space SMC and mixed-domain MALA carry a copy of all N particles along this bridge. Here mixed-domain MALA is the MALA kernel of Section 3.2 applied on the mixed domain (28): the Langevin drift and noise act on all coordinates, each proposal is wrapped back into $\mathcal{Q}$, and the acceptance ratio uses the wrapped normal proposal density on the periodic block. The resulting sample set is held fixed and subsampled during flow training; its smallest incremental ESS also screens the proposed t_k . In KLXX, a second sample set is built by applying QT under $U_{t_k, t_{k-1}}$ to N fresh source draws. Thus the molecular implementation uses a fixed SMC target set and a fresh source QT set, rather than the per-stage target surrogate and validation-seeded QT in Section 4.

After training, the candidate flow and identity proposals are evaluated on the complete carried set $\mathcal{Y}_{k-1}$. The pair with larger ESS over the validation set is denoted by $(\hat{\mathcal{Y}}_k, w_k)$; ties retain the identity. Reweighting, resampling, and MALA under $U_{t_k, t_{k-1}}$ produce $\mathcal{Y}_k^-$. The second arrow holds $t = t_k$ fixed and advances the regularization coordinate from t_{k-1} to t_k . Its exact log weight is

$$\log w_k^{\text{sharp}}(q) = U_{t_k, t_{k-1}}(q) - U_{t_k, t_k}(q) = t_k \left[U^{\{\rho_{t_{k-1}}\}}(q) - U^{\{\rho_{t_k}\}}(q) \right]. \quad (35)$$

Sharpening is accepted only when its ESS over the validation set clears the stage threshold. The samples are then resampled and rejuvenated by MALA under U_{t_k, t_k} . A failed proposal or sharpening gate shrinks $t_k - t_{k-1}$ and retries. The selected map advances the source-to-target interpolation; the stochastic sharpening step changes the molecular potential.

Temperature-annealed Boltzmann generators (TA-BG) (Schopmans and Friederich, 2025) address the same barrier problem through a different continuation coordinate. TA-BG first trains a flow by reverse KL at an elevated temperature, where metastable basins are more connected, and then lowers the temperature through reweighting, resampling, and forward KL updates until the physical target temperature is reached. Sharpening instead keeps the thermodynamic temperature fixed and anneals the surrogate interaction core: the energy threshold e and pair-distance floor r in $U^{\{\rho_s\}}$ move from a softer setting toward the reported endpoint, with exact incremental reweighting and MALA between settings. As a continuation mechanism, this is a stronger intervention at the singular core than temperature scaling alone: changing temperature rescales the physical energy but preserves its divergent short-range form, whereas ρ_s directly changes the regular and exception Coulomb/Lennard–Jones pair floor and the high-energy tail. TA-BG likewise uses an energy regularizer for stable training, but its annealed variable is temperature rather than the regularization setting. This stronger intervention also has a drawback: before the endpoint, the sharpening states are algorithmic surrogate ensembles rather than physical molecular systems because their interaction laws have been modified. By contrast, the TA-BG intermediates remain canonical distributions of the same physical potential at explicitly defined elevated temperatures and can therefore serve as physically interpretable high-temperature reference states. The molecular results below support sharpening as an effective continuation mechanism, but we do not yet have a matched numerical comparison with TA-BG under common targets, force fields, architectures, and energy-evaluation budgets. We leave that comparison to future work.

We compare three generators. ID performs both stochastic arrows without training a flow. KL trains the forward KL and then selects the trained or identity proposal. KLXX uses the same selection rule with log-ratio variation. All three methods retain the same SMC selection, reweighting, resampling, and MALA steps. Consistently with (23), the molecular propagation factor is

$$F = \prod_{k=1}^K \text{ESS}(w_k)^{-1} \text{ESS}(w_k^{\text{sharp}})^{-1}, \quad (36)$$

where $w_k^{\text{sharp}} \equiv 1$ on a fixed regularization path. As in (23), F summarizes incremental weight degeneracy; it is not a fresh source-to-endpoint ESS or an endpoint error estimator.

6.3 Alkane family results

We test methane through n -hexane at 300 K, with mixed-coordinate dimensions 9, 18, $\dots$, 54. Every configuration uses one seed and reaches $t = 1$. All endpoint sample sets target $(e, r) = (100, 0.15)$. Fixed-regularization runs start and remain at this endpoint. Sharpening runs begin at (50, 0.15) for propane, (50, 0.20) for butane and pentane, and (50, 0.25) for hexane, then advance linearly to the common endpoint. The three methods use method-specific adaptive stage schedules, so stage counts and recorded wall times are descriptive rather than fixed-compute comparisons. The reported molecular wall times were recorded on a single NVIDIA GeForce RTX 5090 GPU.

All molecular flows use six mixed spline transforms, 32 bins, conditioner widths (256, 256), spline bound 8, and minimum slope 10^{-3} , and are initialized at the identity. The validation and batch sizes (N, B) are (100000, 5000) for methane, (120000, 6000) for ethane, (160000, 8000) for propane, and (200000, 10000) for butane through hexane. Each accepted flow stage uses 500 Adam steps for methane through propane, 400 for butane, and 300 for pentane and hexane, with learning rate 10^{-3} and 25 warmup steps. Potential-space SMC uses eight annealing levels; mixed-domain MALA uses 100 steps of size 10^{-3} . QT uses melt scale 1, 200 L-BFGS steps with initial step 10^{-2} , and the complete molecular QT set. The safe start is 0.25 for methane and ethane and 0.1 for the larger molecules. The validation gate is 0.4 throughout; the SMC gate is 0.3 for methane and ethane, 0.5 for propane, and 0.8 for butane through hexane.

molecule (d)	start ρ_0	ID			KL			KLXX		
		F	stages	time	F	stages	time	F	stages	time
methane (9)	(100, 0.15)	14.69	8	1.4	1.416	3	1.2	1.053	3	2.4
ethane (18)	(100, 0.15)	62.98	9	3.5	6.349	3	1.8	1.423	3	4.1
propane (27)	(100, 0.15)	182.8	11	7.7	181.8	11	13.4	7.738	7	23.0
	(50, 0.15)	95.14	10	6.9	95.02	10	12.1	7.778	6	18.3
butane (36)	(100, 0.15)	367.3	12	13.3	367.1	12	20.3	28.06	10	49.0
	(50, 0.20)	226.4	10	10.8	224.8	10	14.9	14.91	10	46.1
pentane (45)	(100, 0.15)	733.2	13	19.2	754.7	13	24.6	82.71	12	60.4
	(50, 0.20)	363.6	10	13.9	370.2	10	17.2	48.83	9	38.5
hexane (54)	(100, 0.15)	2043	13	23.3	2077	13	29.0	1026	12	69.9
	(50, 0.25)	910.5	10	17.4	891.5	10	22.7	201.5	10	60.1

Table 6: Alkane Boltzmann generator training results, including propagation factor F from (36), the number of accepted stages, and total wall time in minutes. ID applies the identity map at every stage, whereas KL and KLXX may select a trained flow when it improves validation ESS. Each row reports one seed and one regularization path; bold marks the lowest F among the three methods for that path. Every path ends at (100, 0.15), and rows that start at (100, 0.15) use no sharpening. Wall times are rounded to one decimal, while the time ratios quoted in the text use the unrounded values.

Within the single-seed runs, KLXX has the smallest factor in all ten matched settings. KL selects trained maps through methane and part of ethane. From propane onward, every accepted KL stage selects the identity proposal, and its factor is close to the ID result. KLXX selects the trained map at every accepted stage through pentane. At hexane, it selects trained maps in 6 of 12 fixed-path stages and 8 of 10 sharpening stages.

The difference between the fixed and sharpening paths increases with molecule size in these runs. The propane KLXX factors are nearly equal, whereas the fixed-path factor divided by the sharpening-path factor is 1.88, 1.69, and 5.09 for butane, pentane, and hexane. The corresponding KLXX sharpening runs take 3.10, 2.24, and 2.65 times the recorded KL wall time. Thus the KLXX sharpening paths have the lowest factor in the larger-molecule comparisons, with additional QT, reweighting, and flow-training cost.

The runs stop at the finite regularization ρ_1 rather than at the formal physical limit. For endpoint samples $q_1, \dots, q_N$, define the endpoint-to-physical weights and their normalized endpoint-to-physical regularization

ESS (RESS) by

$$w_i^{\text{phys}} := \exp \left[U^{\{\rho_1\}}(q_i) - U(q_i) \right], \quad \text{RESS} := \text{ESS}(w^{\text{phys}}). \quad (37)$$

The endpoint-to-physical weights are nearly uniform on the final KLXX sample sets: $1 - \text{RESS} = 0$ numerically from methane through butane, 9.5×10^{-6} for pentane, and 2.3×10^{-5} for hexane. These values measure overlap on the sampled sets; they do not prove global equality between $U^{\{\rho_1\}}$ and U .

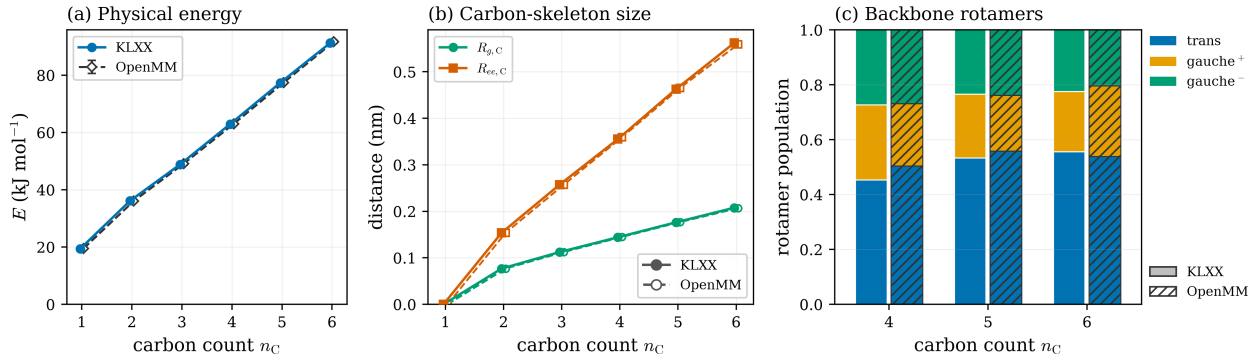


Figure 9: Macroscopic observables from the smallest-factor KLXX sample set for each alkane and an independent physical-potential reference. Fixed-path KLXX is used for methane through propane and sharpening KLXX for butane through hexane. (a) Physical energy evaluated under the exact, unregularized potential. (b) Carbon-skeleton radius of gyration $R_{g,C}$ and terminal-carbon distance $R_{ee,C}$. (c) Pooled backbone trans and gauche fractions. Filled solid curves and unhatched bars denote KLXX; open dashed curves and hatched bars denote the reference.

The KLXX curves are unweighted summaries of the finite-endpoint sample sets; the physical energy E is evaluated only as an observable, without endpoint-to-physical resampling. The physical-potential reference uses two seeds and a 0.25 fs timestep. Methane through propane use 300 K Langevin trajectories; butane through hexane use 300–800 K replica exchange to mix the backbone rotamers. Across the six molecules, the maximum relative difference between KLXX and reference physical-energy means is 0.644%. The largest relative difference among the nonzero carbon radius-of-gyration and terminal-distance means is 1.058%, and the largest absolute rotamer-fraction difference is 0.051. These results show agreement for the selected energy, size, and rotamer summaries across the C_1 – C_6 series. They do not establish equality of the complete distributions.

All thirty ID, KL, and KLXX run manifests used in Table 6 are complete, but each configuration has one training seed. The reference frames are correlated, and the two-seed ranges are descriptive rather than framewise confidence intervals. The table therefore compares the recorded runs; it does not estimate run-to-run variation or distance between the complete equilibrium measures.

6.4 Additional achiral molecule results

We next test NMA, glycerol, and neutral diethanolamine at 300 K, with mixed-coordinate dimensions 30, 36, and 48, respectively. NMA uses ff96/OBC1/ACE, while glycerol and neutral diethanolamine use GAFF2/AM1-BCC with OBC1/ACE. Each method uses one seed, a validation set of size $N = 200000$, and batch size $B = 10000$. The flow, SMC, MALA, and QT settings match the larger-alkane configuration above. NMA and glycerol use 400 Adam steps per accepted flow stage, while neutral diethanolamine uses 300. All paths begin at $\rho_0 = (50, 0.20)$; NMA ends at $\rho_1 = (100, 0.15)$, and the other two molecules end at $\rho_1 = (100, 0.10)$. The two losses use method-specific adaptive stage schedules, so the stage counts and recorded wall times are descriptive rather than fixed-compute comparisons.

molecule (d)	start ρ_0	end ρ_1	KL+X $_{\mu}$			KL+X $_{\mu}$ +X $_{(\hat{\mu}+\bar{\nu})/2}$		
			F	stages	time	F	stages	time
NMA (30)	(50, 0.20)	(100, 0.15)	8.861	6	6.0	4.763	5	12.1
glycerol (36)	(50, 0.20)	(100, 0.10)	604.9	9	16.5	67.69	8	35.8
neutral diethanolamine (48)	(50, 0.20)	(100, 0.10)	1423	12	23.1	471.4	9	43.1

Table 7: Boltzmann generator training results for the additional achiral molecules, including propagation factor F from (36), the number of accepted stages, and total wall time in minutes. The two method groups use the full losses KL+X $_{\mu}$ and KL+X $_{\mu}$ +X $_{(\hat{\mu}+\bar{\nu})/2}$, respectively. Each row reports one seed and one sharpening path; bold marks the smaller F for that molecule. Wall times are rounded to one decimal, while the time ratios quoted in the text use the unrounded values.

Comparing KL+X $_{\mu}$ with KL+X $_{\mu}$ +X $_{(\hat{\mu}+\bar{\nu})/2}$, the latter has the smaller propagation factor for all three molecules. Relative to KL+X $_{\mu}$, it reduces F by factors of 1.86, 8.94, and 3.02 for NMA, glycerol, and neutral diethanolamine, respectively, and reaches $t = 1$ in fewer accepted stages in every case. The corresponding KL+X $_{\mu}$ +X $_{(\hat{\mu}+\bar{\nu})/2}$ runs require 2.03, 2.16, and 1.86 times the recorded wall time. Thus the factor reduction is accompanied by the additional QT and optimization work in KL+X $_{\mu}$ +X $_{(\hat{\mu}+\bar{\nu})/2}$.

Figure 10 resolves each factor into its accepted-stage contributions. Because the two losses use method-specific adaptive stage schedules, the individual arcs are descriptive rather than transition-matched comparisons. Sharpening can be the tighter operation at intermediate stages, so the total factor includes both ESS values reported on each arc.

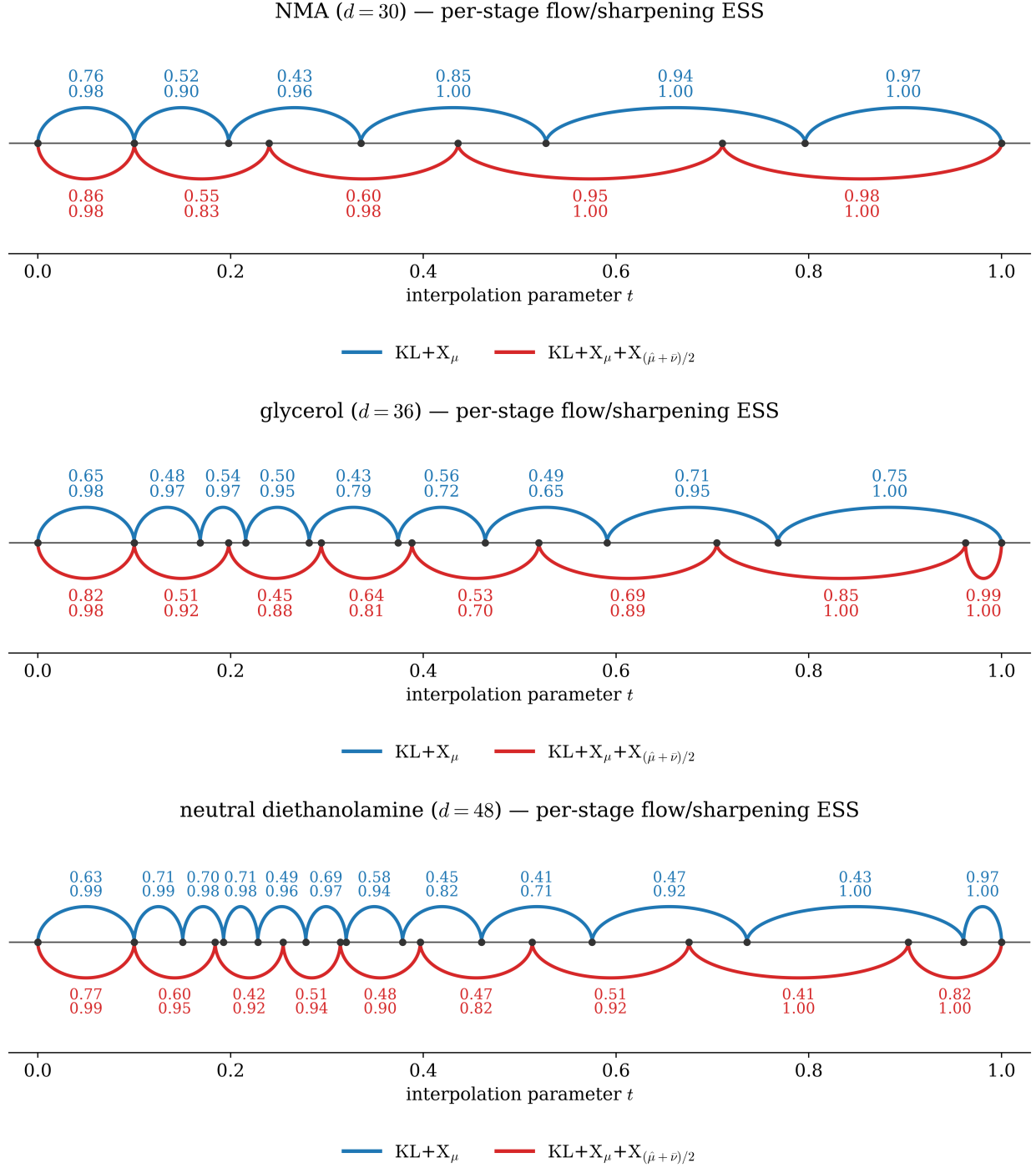


Figure 10: Per-stage flow and sharpening ESS for NMA (top), glycerol (middle), and neutral diethanolamine (bottom). Each arc spans one accepted transition $t_{k-1} \rightarrow t_k$. Within each stacked annotation, the first row is flow ESS and the second row is sharpening ESS. Blue upper arcs show $\text{KL} + \text{X}_\mu$; red lower arcs show $\text{KL} + \text{X}_\mu + \text{X}_{(\hat{\mu} + \hat{\nu})/2}$. The horizontal coordinate is the source-to-target interpolation parameter $t \in [0, 1]$.

The final regularized samples are compared with saved 300 K OpenMM references in Figure 11. The dashed blue curves are plotted above the solid red curves so that close agreement between samples from the two losses remains visible. The selected torsions resolve the NMA amide trans basin and the trans/gauche wells of glycerol and neutral diethanolamine.

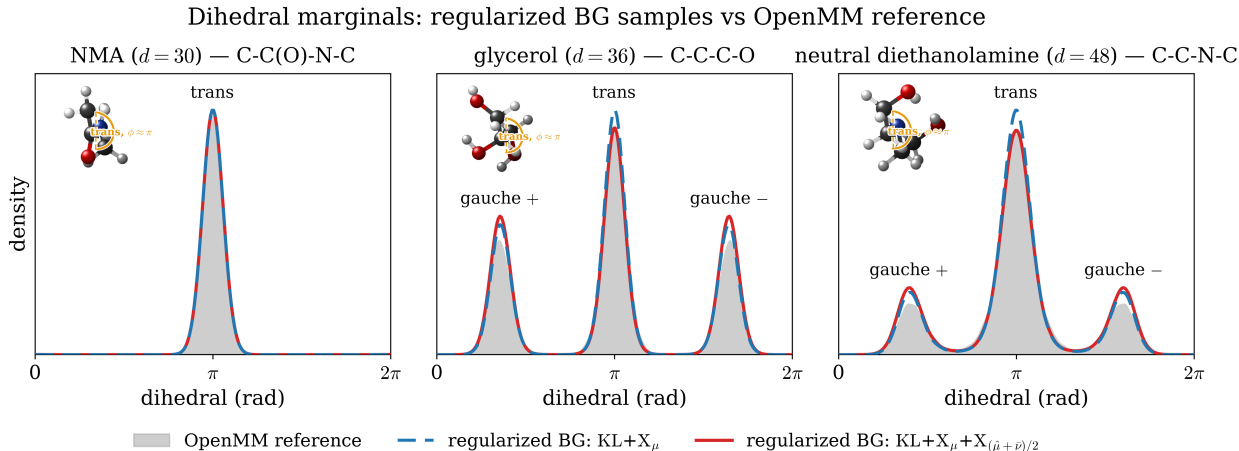


Figure 11: Selected dihedral marginals from regularized Boltzmann generator samples and OpenMM references for NMA, glycerol, and neutral diethanolamine. Gray denotes the OpenMM reference; dashed blue denotes $\text{KL}+X_\mu$, and solid red denotes $\text{KL}+X_\mu+X_{(\hat{\mu}+\hat{\nu})/2}$. The upper-left insets show representative trans conformers. The NMA reference is a saved trajectory initialized in the trans basin, whereas the glycerol and neutral diethanolamine references pool two seeds. The NMA panel is therefore a trans-basin comparison rather than an estimate of the equilibrium cis/trans populations.

The curves overlap closely on these selected one-dimensional marginals, but this diagnostic does not determine the complete molecular distributions. In particular, it does not resolve the NMA cis/trans weights, joint glycerol or neutral diethanolamine rotamers, or unrestricted signed-volume statistics. The reference frames are also correlated, and every generator configuration has one training seed.

6.5 Chiral molecule results

We report completed runs for alanine dipeptide restricted to its L form, (2*R*,3*R*)-2,3-butanediol, and *N*-acetyl-L-proline *N*-methylester (Ac-Pro-NHMe). Only the full KLXX loss $\text{KL}+X_\mu+X_{(\hat{\mu}+\hat{\nu})/2}$ is evaluated, so this subsection contains no within-target loss comparison. Every run uses seed 0, 250 Adam steps per accepted flow stage, an eight-level SMC ladder, and 100 mixed-domain MALA steps. The validation set and batch size pairs are $3.0 \times 10^5/1.5 \times 10^4$, $2.4 \times 10^5/1.2 \times 10^4$, and $3.2 \times 10^5/1.6 \times 10^4$, respectively. The three sharpening paths and their persisted stage summaries are reported in Table 8. After training, each saved stage flow is replayed on a larger particle set without any further optimizer update, giving 10^7 persisted inference samples for alanine dipeptide and 2,3-butanediol and 1.5×10^7 for Ac-Pro-NHMe. The surfaces below use those inference sets; the endpoint-to-physical weights on them are nearly uniform, with RESS equal to 1.000000, 0.999981, and 1.000000, respectively, so the regularized endpoint and the physical potential are indistinguishable on the sampled region.

The two costs are deliberately asymmetric. Training never carries more than 3.2×10^5 samples, and the wall times in Table 8 are correspondingly modest; the large sample sets appear only afterwards, at 10^7 or more, and are produced by replaying frozen flows with no gradient, no optimizer state, and no acceptance gate. The replay is nonetheless the more expensive half in absolute terms, mainly because the NSF inverse is costly: sampling evaluates G^{-1} , which is the expensive direction for this flow structure. Cost therefore sits where it is easiest to pay, since the replay is a fixed number of independent map evaluations and parallelizes across devices, whereas the training schedule is sequential in the stage index and the reference simulations of Section 6.3 must advance their trajectories sequentially in time.

molecule (d)	start ρ_0	end ρ_1	KL+X $_{\mu}$ +X $_{(\hat{\mu}+\hat{\nu})/2}$		
			F	stages	time (min)
alanine dipeptide (60)	(50, 0.25)	(125, 0.10)	111.2	10	72.7
(2 <i>R</i> , 3 <i>R</i>)-2,3-butanediol (42)	(50, 0.20)	(100, 0.10)	22.66	8	35.8
Ac-Pro-NHMe (72)	(50, 0.25)	(150, 0.10)	812.1	11	124.7

Table 8: KLXX-only Boltzmann generator results for the completed chiral targets, including propagation factor F from (36), the number of accepted stages, and total persisted stage time in minutes. The method uses the full loss $\text{KL} + X_{\mu} + X_{(\hat{\mu} + \hat{\nu})/2}$. Each row reports one seed and one sharpening path; no alternative loss was run for these targets.

All three manifests are complete and reach $t = 1$. The propagation factors multiply the reciprocals of the flow and sharpening ESS values over accepted stages, as in the earlier molecular tables. Because the rows correspond to different molecules, validation sizes, and regularization paths, their factors are stage-propagation diagnostics rather than a cross-molecule ranking.

Figure 12 shows one representative final-stage conformation for each completed target. The plotted structure is an actual validation sample selected near the most populated two-dihedral basin. Gold rings mark the fixed stereocenters: the *S* alanine C_{α} , both *R* centers of 2,3-butanediol, and the *S* proline C_{α} .

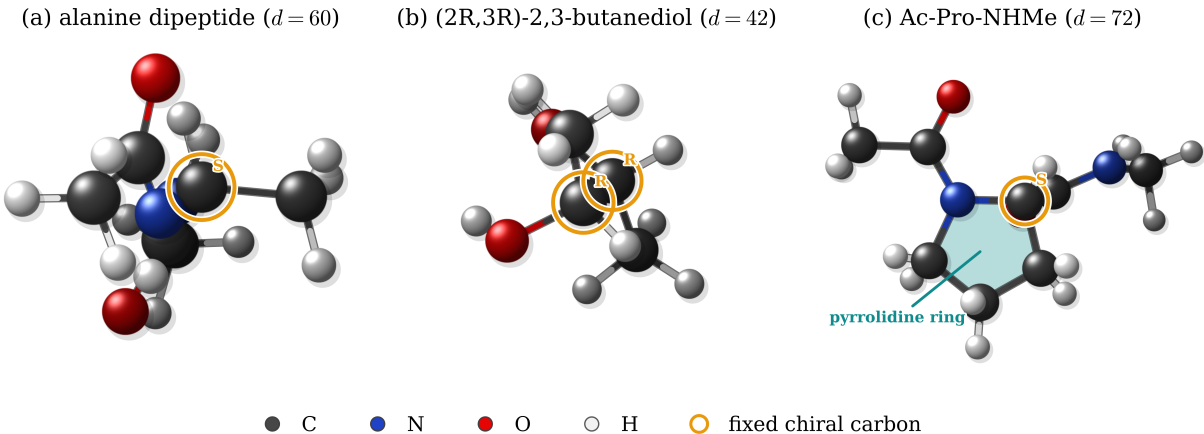


Figure 12: Representative final-stage conformations for the three completed chiral targets. From left to right: L-alanine dipeptide, (2*R*,3*R*)-2,3-butanediol, and Ac-Pro-NHMe. Gold rings and the adjacent *R/S* labels identify every fixed stereocenter; the atom colors follow the CPK convention shown in the legend. Each structure is the saved validation sample nearest the modal center of a molecule-specific two-dihedral histogram.

For alanine dipeptide, Figure 13 tracks the Ramachandran surface along four accepted sharpening stages. Each panel uses 10^7 persisted inference samples generated without additional training. The panels correspond to different regularized potentials, so the visible change is a path diagnostic rather than repeated estimation of one fixed target.

Two of the three chiral targets carry an independent equilibrium reference. For alanine dipeptide the reference is published: the implicit-solvent trajectory data released with the FAB study (Midgley et al., 2022) were generated for the same Hamiltonian and are used here as ground truth. Table 9 compares the final KLXX stage against it. Both sets contain 10^7 samples, are binned on the same 100×100 grid over (ϕ, ψ) , and are smoothed identically, and each surface is normalized to its own most populated bin, so only per-method values are quoted; a difference between two such surfaces would carry an arbitrary additive offset.

The two sets agree throughout. The three basin populations differ by at most 0.0018, including the

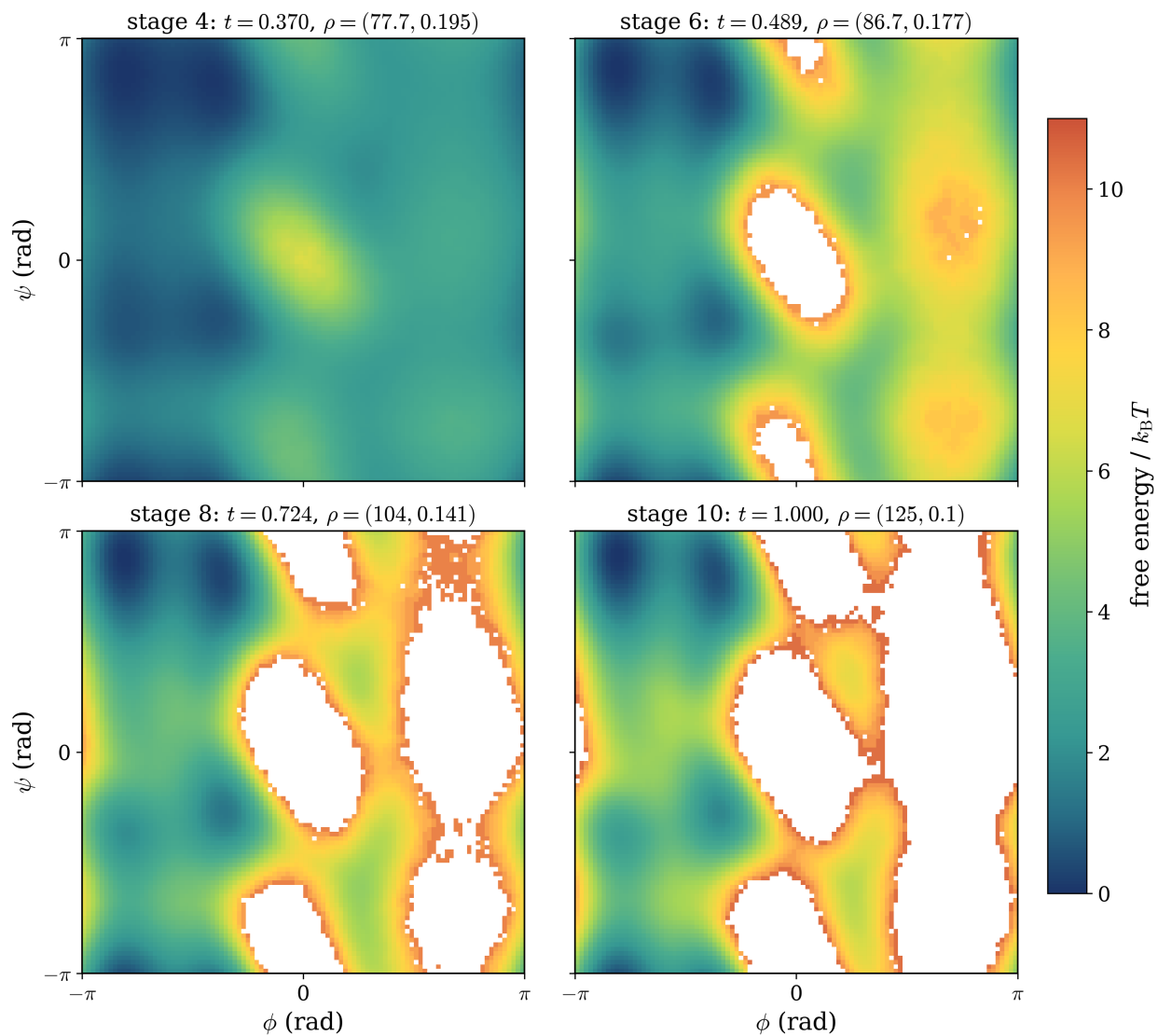


Figure 13: Alanine dipeptide Ramachandran free energy surfaces at accepted stages 4, 6, 8, and 10. Each panel contains 10^7 persisted inference samples obtained by generating pushforward samples with the corresponding saved flow and then applying resampling, rejuvenation, sharpening, and post-sharpen rejuvenation. The panel titles give the interpolation parameter and regularization value; stage 10 reaches $t = 1$ and $\rho = (125, 0.10)$. A common colorbar reports free energy relative to the observed minimum in units of $k_B T$.

sample set	backbone basin population		
	α_L	β/PPII	α_R
KL+X $_{\mu}$ +X $_{(\hat{\mu}+\bar{\nu})/2}$	0.0031	0.7958	0.2012
ground truth	0.0033	0.7940	0.2027
sample set	mean free energy over reference-classified bins		
	low	high	all
KL+X $_{\mu}$ +X $_{(\hat{\mu}+\bar{\nu})/2}$	1.230	6.323	5.379
ground truth	1.230	6.293	5.355

Table 9: Alanine dipeptide at the final accepted stage against ground truth, the implicit-solvent reference data of Midgley et al. (2022), both with 10^7 samples. Basins split the (ϕ, ψ) plane by $\phi > 0$ for α_L and, for $\phi \leq 0$, by the sign of ψ . Bins are classified by the reference surface as low below $2k_B T$ and high above $3k_B T$, and restricted to bins finite in both surfaces; entries are mean free energies in units of $k_B T$. Both surfaces reach the same maximum of $10.40 k_B T$.

sparsely populated α_L region that carries only 0.3% of the total, and the mean free energies agree to $0.03 k_B T$ in every band. The alanine dipeptide comparison is therefore highly consistent; a single target does not by itself establish accuracy across systems.

Figure 14 compares the accepted 2,3-butanediol stage nearest $t = 0.5$ with the final stage. Both rows resolve the heavy-atom rotamers and hydroxyl orientations; each diagnostic shares one color scale across the two stages, while the two columns retain separate colorbars.

2,3-butanediol carries an independent equilibrium reference of our own: a PT run on the unregularized physical potential at 300 K, six replicas on a geometric 300–800 K grid, retaining 10^5 frames over 50.0 ns after a discarded 0.5 ns equilibration, with a mean per-pair exchange probability of 0.529 and 2.2×10^4 transitions among the three central rotamers. Table 10 compares it with the final KLXX stage on the same 100×100 torsion grid with the same smoothing.

The two sample sets differ in size by two orders of magnitude, and the two halves of the table are read accordingly. A basin population and a mean over occupied bins are ratios, estimated without bias at either size, so both are quoted from the full 10^7 inference set. A maximum free energy is not such a quantity: each surface is normalized to its own most populated bin, so the deepest value it can resolve is set by the bin holding a single sample and grows as $\log N$. Maxima are therefore quoted only at matched count, with the same inference set thinned to the reference’s 10^5 .

sample set	central C-C-C-C rotamer population		
	gauche−	gauche+	trans
KL+X $_{\mu}$ +X $_{(\hat{\mu}+\bar{\nu})/2}$	0.1486	0.1861	0.6652
OpenMM reference	0.1425	0.1726	0.6849
sample set	mean free energy over reference-classified bins		
	low	high	all
KL+X $_{\mu}$ +X $_{(\hat{\mu}+\bar{\nu})/2}$	1.229	5.126	4.137
OpenMM reference	1.259	5.353	4.309

Table 10: (2*R*, 3*R*)-2,3-butanediol at the final accepted stage against an independent OpenMM PT reference, with 10^7 inference samples and 10^5 reference frames. Rotamers split the central C-C-C-C torsion at its eclipsed barriers, trans for $|\phi| \geq 2\pi/3$ and one gauche well on either side of $\phi = 0$. Bins are classified by the reference surface as low below $2k_B T$ and high above $3k_B T$, and restricted to bins finite in both surfaces; entries are mean free energies in units of $k_B T$ on the heavy-atom torsions. Maximum free energies are comparable only at matched count: with the inference set thinned to 10^5 , they are 7.52 against 7.54 on the heavy-atom torsions and 4.04 against 4.04 on the hydroxyl pair.

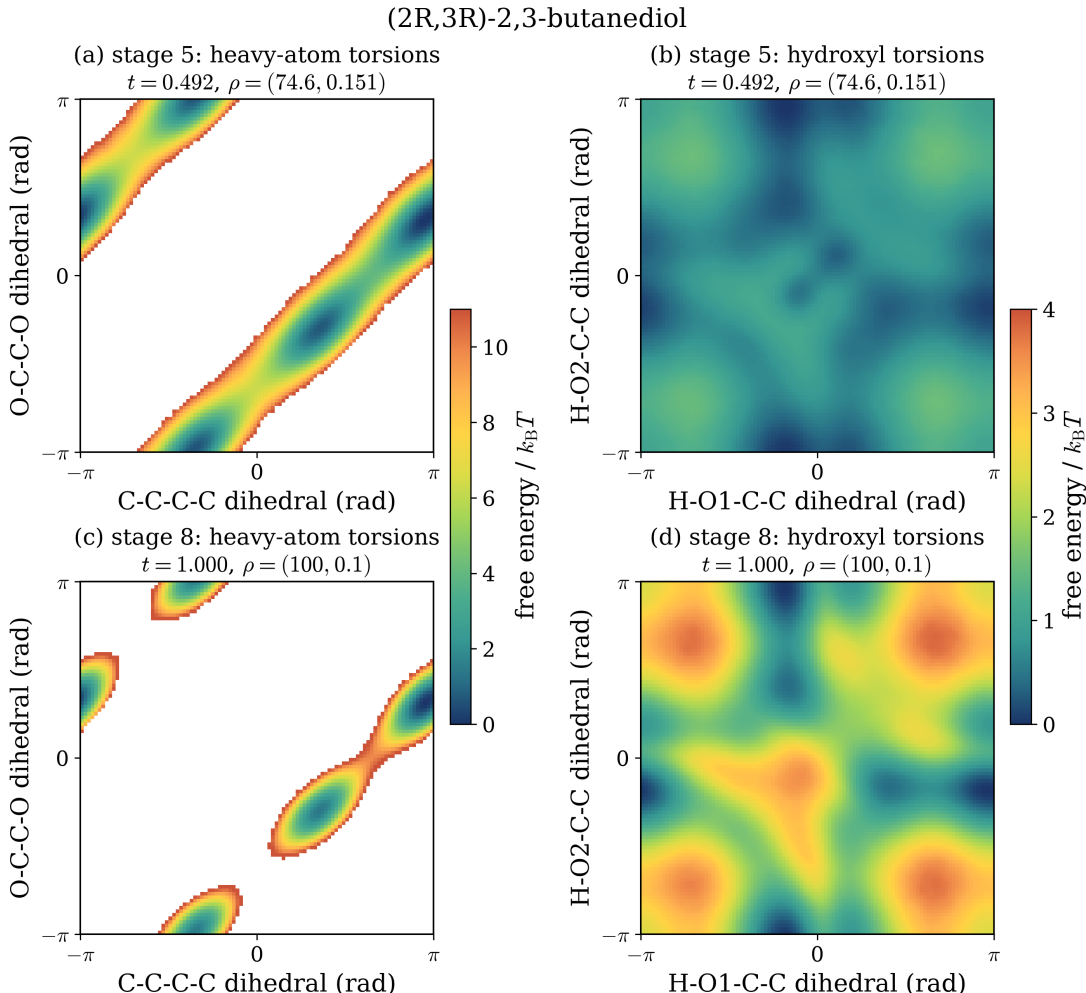


Figure 14: Conformational free energy surfaces for (2R,3R)-2,3-butanediol from 10^7 persisted inference samples per stage. The upper row is stage 5, the accepted stage nearest $t = 0.5$, at $t = 0.492$ and $\rho = (74.6, 0.151)$; the lower row is the final stage 8 at $t = 1$ and $\rho = (100, 0.10)$. The left column shows the C-C-C-C and O-C-C-O heavy-atom torsions, and the right column shows the two hydroxyl-orientation torsions. Each column shares its scale across stages, and the two diagnostics have separate colorbars in units of $k_B T$.

The two sets agree closely. The rotamer populations differ by at most 0.020, in the dominant trans well, and by 0.014 and 0.006 in the two gauche wells; the mean free energies differ by at most $0.23 k_B T$; and at matched sample count the maxima differ by $0.02 k_B T$ and $0.00 k_B T$. The residual trans difference is larger than the reference’s own sampling error, which the rotamer transition count places near 0.003, so it is not attributable to reference noise alone.

Ac-Pro-NHMe adds a ring degree of freedom and a cis/trans peptide torsion. Figure 15 compares the accepted stage nearest $t = 0.5$ with the final stage in both the backbone and joint ACE-PRO torsion/ring-pucker coordinates.

These four figures diagnose the regularized KLXX sample sets and their retained stereochemical components. Alanine dipeptide is additionally checked against ground truth in Table 9, and 2,3-butanediol against our own reference in Table 10. What this subsection does not provide is a matched loss comparison, an independent equilibrium reference for Ac-Pro-NHMe, or uncertainty over independently trained generators.

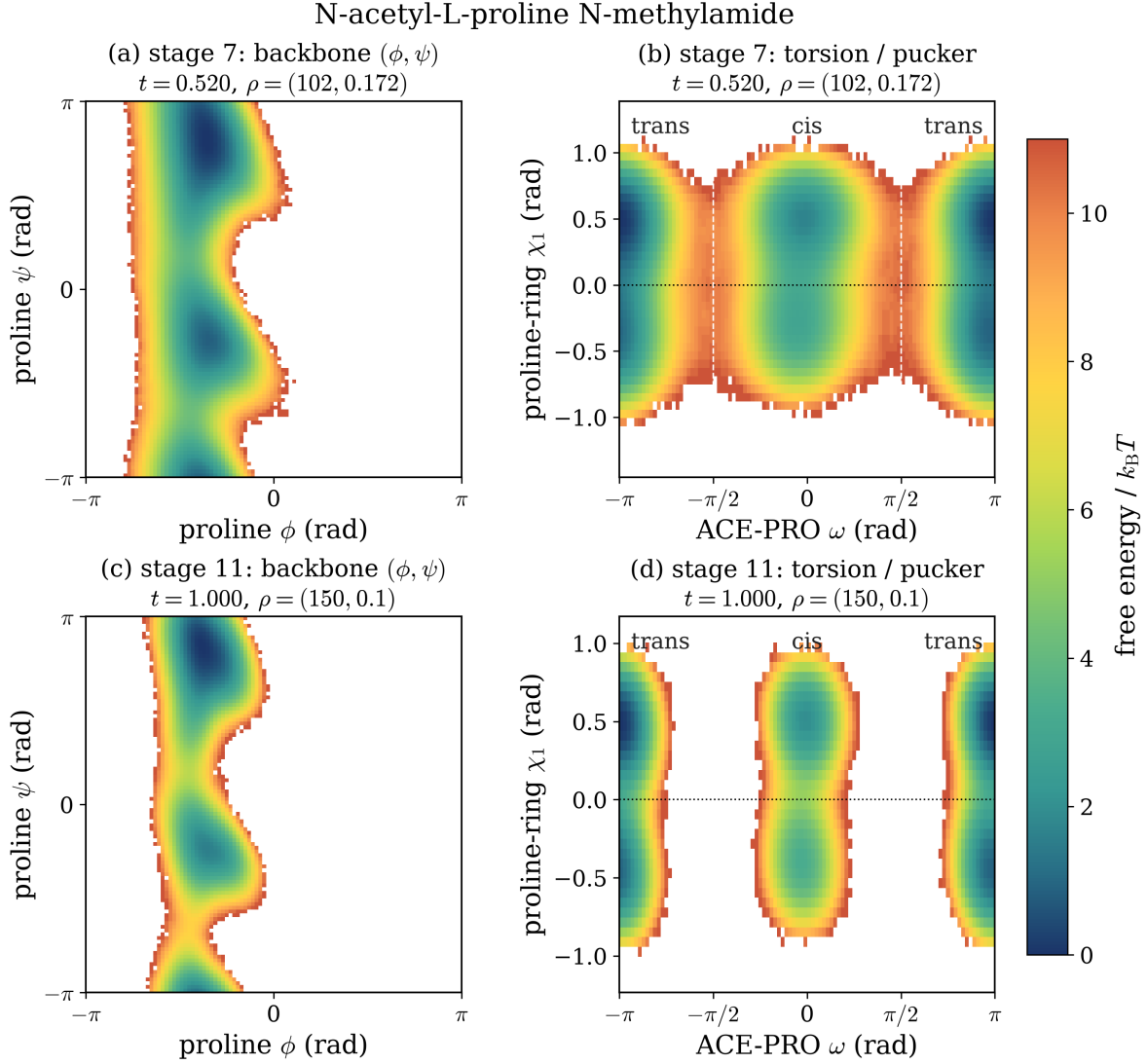


Figure 15: Conformational free energy surfaces for Ac-Pro-NHMe from 1.5×10^7 persisted inference samples per stage. The upper row is stage 7, the accepted stage nearest $t = 0.5$, at $t = 0.520$ and $\rho = (102, 0.172)$; the lower row is the final stage 11 at $t = 1$ and $\rho = (150, 0.10)$. The left column shows the proline backbone (ϕ, ψ) surface, and the right column shows the joint ACE-PRO peptide torsion ω and proline-ring pucker χ_1 , with the cis and trans sectors marked. All four panels share one free energy scale in units of $k_B T$.

7 Conclusions

Forward KL is mass-covering for exact distributions, but its sampled loss sees only the target regions supplied to it. This gap can produce a fake ESS, transfer surrogate error to the flow, and leave target-poor proposal regions weakly constrained. Log-ratio variation addresses these failures by comparing the exact log-density ratio at pairs of samples. The weighting measure determines which regions are compared: target samples measure density variation on reached target regions, QT samples add candidate modes, and pushforward samples expose regions occupied by the proposal. KLXX combines these comparisons while retaining the exact target as a minimizer and the flow as a tractable importance proposal.

The theory establishes a narrower statement. For the FR gradient flow of forward KL with X_μ , the regularizer adds a nonpositive term to the instantaneous KL dissipation. For a fixed smooth surrogate and $\lambda < \frac{1}{2}$, it also subtracts a nonnegative discrepancy term from the stationary-error bound. This does not prove finite-step behavior for full KLXX, Adam, or the experimental value $\lambda = 1$.

The numerical results provide the remaining evidence. The 2D and ϕ^4 ablations are consistent with the roles of the target, QT, and pushforward samples; the product test shows higher validation ESS for every tested objective with log-ratio variation at all sixteen tested dimensions. On the shared clock stage schedules, KLXX has higher validation ESS on all 28 shared stages and a smaller propagation factor at every reported batch size. In the single-seed alkane runs, KLXX has the smallest factor in all ten matched settings, while forward KL approaches the identity baseline as molecule size grows. The sharpening paths have lower KLXX factors for the larger alkanes. For NMA, glycerol, and neutral diethanolamine, KLXX also has a smaller factor than $\text{KL}+X_\mu$ and reaches the endpoint in fewer accepted stages. The three chiral KLXX runs reach $t = 1$, retain their specified stereochemical components, and populate multiple torsional basins in the reported conformational diagnostics. For alanine dipeptide, the final sample set matches the published equilibrium reference of Midgley et al. (2022) at 10^7 samples per side: the three backbone basin populations differ by at most 0.0018, including the 0.3% α_L basin, and the mean free energies agree to $0.03 k_B T$ in every band. Selected alkane observables and achiral dihedral marginals agree with independent physical-potential simulations on the reported diagnostics. These findings are consistent with an explanation based on additional support information; they are not a theorem of universal dominance or a fixed-compute comparison.

The clock and molecular configurations require additional training seeds to measure run-to-run variation. The finite alkane endpoints have nearly uniform endpoint-to-physical weights on the reported sample sets, but a continuation to U would remove the remaining regularization. The additional achiral figures report selected one-dimensional marginals and do not determine joint conformational weights. Among the chiral results, alanine dipeptide is checked against published ground truth and 2,3-butanediol against a reference of our own; Ac-Pro-NHMe lacks a matched independent MD reference, and no alternative-loss run was made for any of the three. Larger molecules and explicit solvent remain outside the completed results.

Acknowledgments

The normalizing flow computations use the JAX backend through two Python packages maintained by Xuda Ye. The packages `jflows` and `jflows-md` can be installed directly from PyPI with:

```
pip install jflows jflows-md
```

Molecular benchmark data were generated with OpenMM and AmberTools, and the alanine dipeptide reference data were taken from the FAB study (Midgley et al., 2022). The source files are available at:

```
https://github.com/xuda-ye-math/KLXX
```

The GPU-accelerated runs used one MSI GeForce RTX 5090 32G Gaming Trio OC with 32 GB of VRAM. The card ran at a 600 W power limit with clock offsets of +250 MHz on the core and +2000 MHz on the memory. All reported wall times correspond to this overclocked configuration.

A Fisher–Rao gradient flow with log-ratio variation

Section 2 states the main gradient flow results used in the paper. This appendix supplies their derivations and proofs on the space of densities. We view ν as a generic probability density evolving along the gradient flow of the loss under an appropriate Riemannian metric on the space of densities (Ambrosio et al., 2008; Villani, 2009) and ask how fast ν_t converges to the target μ in $\text{KL}(\mu\|\nu_t)$.

At ties we set $\text{sgn}(0) = 0$. Throughout this appendix, the densities are assumed positive and sufficiently regular for the displayed gradient calculations.

In this section we restrict to forward KL with the target-weighted regularizer X_μ , the log-ratio variation at weighting density $\omega = \mu$,

$$\mathcal{L}[\nu] = \text{KL}(\mu\|\nu) + \lambda X_\mu(\mu\|\nu) = \int_{\mathbb{R}^d} \mu(x) \log \frac{\mu(x)}{\nu(x)} dx + \lambda \iint_{\mathbb{R}^d \times \mathbb{R}^d} \mu(x) \mu(y) \left| \log \frac{\mu(x)}{\nu(x)} - \log \frac{\mu(y)}{\nu(y)} \right| dx dy, \quad (38)$$

where $\lambda > 0$ is fixed. The calculations below hold along a positive solution of this gradient flow; when $\lambda \leq \frac{1}{2}$, the birth term in (11) is nonnegative and preserves positivity. Throughout this section we abbreviate the importance weight by

$$w(x) := \frac{\mu(x)}{\nu(x)},$$

so that the log-ratio is $z(x) = \log w(x) = \log \mu(x) - \log \nu(x)$, the field z of (6). The importance weight w is distinct from the weighting density ω in X_ω .

The choice of Riemannian metric matters. In the Wasserstein-2 geometry, induced by optimal transport (Jordan et al., 1998; Otto, 2001), the forward KL fails to be displacement convex along Wasserstein-2 geodesics, and the standard Otto–Villani/Bakry–Émery argument (Otto and Villani, 2000; Bakry et al., 2014) used for the reverse KL does not apply. We therefore work in the Fisher–Rao (FR) geometry (Amari, 2016; Bauer et al., 2016). Along the positive flow, the forward KL has an exponential bound and X_μ contributes a nonlocal birth–death term (Lu et al., 2019): the update at x depends on how its importance weight ranks among weights under μ .

Denoting the ν -mean by $\langle \cdot \rangle_\nu := \int (\cdot) \nu dx$, the FR gradient of a functional $\mathcal{F}[\nu]$ and the associated dissipation identity along the FR gradient flow $\partial_t \nu_t = -\text{grad}^{\text{FR}} \mathcal{F}(\nu_t)$ are

$$\text{grad}^{\text{FR}} \mathcal{F}(\nu) = \nu \left(\frac{\delta \mathcal{F}}{\delta \nu} - \left\langle \frac{\delta \mathcal{F}}{\delta \nu} \right\rangle_\nu \right), \quad \frac{d}{dt} \mathcal{F}[\nu_t] = - \int_{\mathbb{R}^d} \left(\frac{\delta \mathcal{F}}{\delta \nu} - \left\langle \frac{\delta \mathcal{F}}{\delta \nu} \right\rangle_{\nu_t} \right)^2 \nu_t dx. \quad (39)$$

The mean-subtraction enforces $\int \text{grad}^{\text{FR}} \mathcal{F} = 0$, so the FR gradient flow conserves probability mass.

A.1 Derivation of X_μ -regularized Fisher–Rao gradient flow

The first variation of the forward KL with respect to ν is

$$\frac{\delta \text{KL}(\mu\|\nu)}{\delta \nu}(x) = -\frac{\mu(x)}{\nu(x)} = -w(x), \quad (40)$$

so by (39), using $\langle w \rangle_\nu = \int_{\mathbb{R}^d} \mu dx = 1$, the FR gradient of the forward KL is

$$\text{grad}^{\text{FR}} [\text{KL}(\mu\|\nu)](x) = \nu(x) - \mu(x). \quad (41)$$

For X_μ , the variation uses $\text{sgn}(0) = 0$ at ties. At an evaluation point r , the variation of the inner log-ratio difference is

$$\frac{\delta}{\delta \nu(r)} [\log w(x) - \log w(y)] = -\frac{\delta(r-x)}{\nu(x)} + \frac{\delta(r-y)}{\nu(y)}.$$

Here δ is the Dirac distribution, so integration against $\delta(r-x)$ evaluates the remaining integrand at $x=r$. Applying this sign convention and then exchanging x and y in the second term gives

$$\begin{aligned} \frac{\delta X_\mu(\mu\|\nu)}{\delta \nu}(r) &= \iint_{\mathbb{R}^d \times \mathbb{R}^d} \mu(x) \mu(y) \text{sgn}(w(x) - w(y)) \left[-\frac{\delta(r-x)}{\nu(x)} + \frac{\delta(r-y)}{\nu(y)} \right] dx dy \\ &= -2w(r) \int_{\mathbb{R}^d} \mu(y) \text{sgn}(w(r) - w(y)) dy. \end{aligned}$$

Hence the variation is

$$\frac{\delta X_\mu(\mu\|\nu)}{\delta\nu}(x) = -2w(x)S(x), \quad S(x) := \mathbb{E}_{y\sim\mu}[\text{sgn}(w(x) - w(y))]. \quad (42)$$

We call S the *rank correlation* of the importance weight under μ : $S(x) > 0$ indicates that $w(x)$ is comparatively large within the distribution of weights, and hence that ν pointwise undercovers μ relative to that distribution; $S(x) < 0$ has the opposite interpretation, and $S \equiv 0$ at $\nu = \mu$. By antisymmetry,

$$\mathbb{E}_\mu[S] = \mathbb{E}_{x,y\sim\mu}[\text{sgn}(w(x) - w(y))] = 0,$$

and the FR gradient of X_μ with respect to ν is

$$\text{grad}^{\text{FR}}[X_\mu(\mu\|\nu)](x) = -2\mu(x)S(x). \quad (43)$$

The FR gradient of $\mathcal{L}[\nu] = \text{KL}(\mu\|\nu) + \lambda X_\mu(\mu\|\nu)$ is the sum of (41) and (43):

$$\text{grad}^{\text{FR}}[\mathcal{L}](x) = \nu(x) - \mu(x) - 2\lambda\mu(x)S(x).$$

This yields the FR gradient flow (11) stated in Section 2. It is a nonlocal partial differential equation (PDE): its right-hand side at x depends on $S_t(x)$, which in turn requires the entire profile of the importance weight μ/ν_t over the support of μ . Mass is conserved along the flow because

$$\frac{d}{dt} \int_{\mathbb{R}^d} \nu_t(x) dx = 2\lambda \int_{\mathbb{R}^d} \mu(x) S_t(x) dx = 0,$$

and the flow is stationary at $\nu_t = \mu$, where $\mu(x)/\nu_t(x) \equiv 1$ and hence $S_t \equiv 0$. When $\lambda = 0$, (11) reduces to the FR gradient flow of forward KL.

A.2 Convergence analysis of the X_μ -regularized flow

To analyze the convergence rate of the regularized flow (11), we take the forward KL $\text{KL}(\mu\|\nu_t)$ as the error metric and differentiate it along the flow:

$$\frac{d}{dt} \text{KL}(\mu\|\nu_t) = - \int_{\mathbb{R}^d} \frac{\mu^2}{\nu_t} (1 + 2\lambda S_t) dx + \int_{\mathbb{R}^d} \mu dx = -\chi^2(\mu\|\nu_t) - 2\lambda \int_{\mathbb{R}^d} \frac{\mu^2}{\nu_t} S_t dx,$$

where the chi-squared divergence is

$$\chi^2(\mu\|\nu) := \int_{\mathbb{R}^d} (w - 1)^2 \nu dx = \int_{\mathbb{R}^d} w^2 \nu dx - 1.$$

The sign of the second integral is settled by the following rank correlation identity.

Lemma 1 *Let μ and ν be positive probability densities for which the displayed integral is finite. Then*

$$\int_{\mathbb{R}^d} \frac{\mu(x)^2}{\nu(x)} S(x) dx = \frac{1}{2} \mathbb{E}_{x,y\sim\mu}[|w(x) - w(y)|] \geq 0,$$

with equality if and only if w is μ -almost-everywhere constant.

Proof. Since $\mu^2/\nu = \mu w$, the definition of S gives

$$\int_{\mathbb{R}^d} \frac{\mu^2}{\nu} S dx = \mathbb{E}_\mu[w(x)S(x)] = \mathbb{E}_{x,y\sim\mu}[w(x) \text{sgn}(w(x) - w(y))].$$

Symmetrizing in $x \leftrightarrow y$ gives

$$\int_{\mathbb{R}^d} \frac{\mu^2}{\nu} S dx = \frac{1}{2} \mathbb{E}_{x,y\sim\mu}[(w(x) - w(y)) \text{sgn}(w(x) - w(y))] = \frac{1}{2} \mathbb{E}_{x,y\sim\mu}[|w(x) - w(y)|].$$

□

Applying Lemma 1 with $\nu = \nu_t$ and $S = S_t$, the dissipation can be rewritten as

$$\frac{d}{dt} \text{KL}(\mu \| \nu_t) = -\chi^2(\mu \| \nu_t) - \lambda \mathbb{E}_{x, y \sim \mu} \left[\left| \frac{\mu(x)}{\nu_t(x)} - \frac{\mu(y)}{\nu_t(y)} \right| \right]. \quad (44)$$

The elementary inequality $w \log w \leq w^2 - w$ applied to $w = \mu/\nu_t$ gives $\chi^2(\mu \| \nu_t) \geq \text{KL}(\mu \| \nu_t)$, hence $\frac{d}{dt} \text{KL}(\mu \| \nu_t) \leq -\text{KL}(\mu \| \nu_t)$, and Grönwall's lemma yields the exponential decay

$$\text{KL}(\mu \| \nu_t) \leq e^{-t} \text{KL}(\mu \| \nu_0), \quad t \geq 0, \quad (45)$$

with no structural assumption on μ : neither log-concavity, nor a spectral gap, nor any growth condition. This contrasts with the Wasserstein-2 geometry, where the forward KL is not displacement convex and an analogous exponential rate is unavailable without further assumptions.

The uniform convergence rate (45) should *not* be read as a blanket statement that minimizing the forward KL is intrinsically faster or more efficient than minimizing the reverse KL. The FR gradient flow is a continuous-time idealization on the space of densities and does not account for sampling cost. In practice, evaluating the forward KL and its gradient requires samples from μ , and producing representative samples in high dimensions is itself difficult. The empirical KLXX objective therefore also evaluates the exact log-ratio on the quench and temper measure $\hat{\mu}$, which is intended to supply support information that may be absent from the target-sample measure. The bound above quantifies convergence given exact target expectations; it does not establish the behavior of this additional empirical term or remove the cost of constructing its samples.

A naive comparison of dissipation rates between forward KL and $\text{KL} + \lambda X_\mu$ would also be misleading, since scaling the loss by a constant is mere time reparametrization with no qualitative gain. The substantive contribution of X_μ is structural rather than rate-based: the term from forward KL depends locally on the value of $\mu(x)/\nu_t(x)$, whereas the rank correlation S_t compares that value with the complete distribution of importance weights under μ . The regularized flow (11) therefore couples the update at x to this rank correlation, which the term from forward KL does not contain.

Two further cautions concern the discrete realization of the flow. First, the time-reparametrization argument cuts in both directions once the optimizer enters. Training follows Adam steps, not the FR gradient flow: the per-coordinate second-moment normalization absorbs a global rescaling of the loss, so a larger gradient buys no speedup by itself, while a regularizer that reshapes the relative magnitudes of the gradient components does change the optimizer's effective time scale. Part of any observed difference in training speed between forward KL and $\text{KL} + \lambda X_\mu$ may therefore be optimizer scaling rather than information geometry, and we claim no universal transient speedup on this basis. Second, the dissipation (44) is a density-space statement, while each gradient step uses a finite, generally biased sample estimate. The staged clock experiment examines this finite-batch setting with a schedule-matched comparison: its quantitative claims use the selected-proposal ESS over the validation set at each stage, not a composed endpoint estimate or a diagnostic computed on a training mini-batch.

Despite these cautions, the density-space flow retains a further monotonicity property: X_μ decreases along the regularized flow. Pairing the variation $\delta X_\mu / \delta \nu = -2wS$ (42) against the flow (11) gives

$$\frac{d}{dt} X_\mu(\mu \| \nu_t) = -2 \int_{\mathbb{R}^d} \frac{\mu^2}{\nu_t} S_t dx - 4\lambda \int_{\mathbb{R}^d} \frac{\mu^2}{\nu_t} S_t^2 dx \leq 0.$$

The first term is nonpositive by Lemma 1 and vanishes only when μ/ν_t is μ -almost-everywhere constant. Hence the inequality is strict away from that case, and X_μ is driven to zero along the flow.

A.3 Asymptotic accuracy for a fixed biased surrogate

The convergence analysis of Section A.2 takes the outer expectation in $\text{KL}(\mu \| \nu) + \lambda X_\mu(\mu \| \nu)$ under the exact target. To isolate surrogate bias, fix a smooth density $\tilde{\mu}$ that does not vary with ν and define

$$\tilde{\mathcal{L}}[\nu] = \int_{\mathbb{R}^d} \tilde{\mu}(x) \log w(x) dx + \lambda \iint_{\mathbb{R}^d \times \mathbb{R}^d} \tilde{\mu}(x) \tilde{\mu}(y) |\log w(x) - \log w(y)| dx dy, \quad (46)$$

where the pointwise ratio $w(x) = \mu(x)/\nu(x)$ is still evaluated against the exact target. Repeating the gradient calculation of Section A.1 with μ replaced by $\tilde{\mu}$ only in the outer integrals gives

$$\text{grad}^{\text{FR}}[\tilde{\mathcal{L}}](x) = \nu(x) - \tilde{\mu}(x)(1 + 2\lambda\tilde{S}(x)), \quad \tilde{S}(x) := \mathbb{E}_{y \sim \tilde{\mu}}[\text{sgn}(w(x) - w(y))].$$

A stationary density ν^* of this gradient flow therefore satisfies

$$\nu^*(x) = \tilde{\mu}(x) \left[1 + 2\lambda \int_{\mathbb{R}^d} \tilde{\mu}(y) \text{sgn}\left(\frac{\mu(x)}{\nu^*(x)} - \frac{\mu(y)}{\nu^*(y)}\right) dy \right]. \quad (47)$$

At $\lambda = 0$, the equation reduces to $\nu^* = \tilde{\mu}$ and the residual error is $\text{KL}(\mu \parallel \tilde{\mu})$. For $\lambda > 0$, the rank correlation $\tilde{S}^*$ enters through $1 + 2\lambda\tilde{S}^*$, raising ν^* relative to $\tilde{\mu}$ where μ/ν^* is comparatively high and lowering it where the ratio is low. For $\lambda < \frac{1}{2}$, (47) is positive because $\tilde{S}^* \in [-1, 1]$. Antisymmetry of the sign also gives

$$\int_{\mathbb{R}^d} \tilde{\mu}(x) \tilde{S}^*(x) dx = \mathbb{E}_{x, y \sim \tilde{\mu}} \left[\text{sgn}\left(\frac{\mu(x)}{\nu^*(x)} - \frac{\mu(y)}{\nu^*(y)}\right) \right] = 0,$$

hence every stationary point $\nu^* = \tilde{\mu}(1 + 2\lambda\tilde{S}^*)$ is a probability density.

We now prove Theorem 1 of Section 2, whose fixed-surrogate loss (14) is the loss (46) derived above.

Proof. Substituting the stationary identity $\nu^* = \tilde{\mu}(1 + 2\lambda\tilde{S}^*)$ into the difference of forward KLs and using $\mu = \nu^* \cdot (\mu/\nu^*) = \tilde{\mu}(1 + 2\lambda\tilde{S}^*)(\mu/\nu^*)$,

$$\text{KL}(\mu \parallel \tilde{\mu}) - \text{KL}(\mu \parallel \nu^*) = \int_{\mathbb{R}^d} \mu(x) \log(1 + 2\lambda\tilde{S}^*(x)) dx = \int_{\mathbb{R}^d} \tilde{\mu}(x) \frac{\mu(x)}{\nu^*(x)} u(x) \log u(x) dx,$$

where $u(x) := 1 + 2\lambda\tilde{S}^*(x) \geq 1 - 2\lambda > 0$, since $\tilde{S}^* \in [-1, 1]$ (a $\tilde{\mu}$ -average of signs) and $\lambda < \frac{1}{2}$. The elementary inequality $u \log u \geq u - 1$ for $u > 0$ then yields

$$\text{KL}(\mu \parallel \tilde{\mu}) - \text{KL}(\mu \parallel \nu^*) \geq 2\lambda \int_{\mathbb{R}^d} \tilde{\mu}(x) \frac{\mu(x)}{\nu^*(x)} \tilde{S}^*(x) dx.$$

Applying the rank correlation identity of Lemma 1 verbatim with μ replaced by $\tilde{\mu}$ in the outer measure,

$$\int_{\mathbb{R}^d} \tilde{\mu}(x) \frac{\mu(x)}{\nu^*(x)} \tilde{S}^*(x) dx = \frac{1}{2} \iint_{\mathbb{R}^d \times \mathbb{R}^d} \tilde{\mu}(x) \tilde{\mu}(y) \left| \frac{\mu(x)}{\nu^*(x)} - \frac{\mu(y)}{\nu^*(y)} \right| dx dy,$$

which gives (15). $\square$

At $\lambda = 0$, the bound is attained by $\nu^* = \tilde{\mu}$. For $0 < \lambda < \frac{1}{2}$, the subtracted discrepancy is nonnegative and is positive unless μ/ν^* is constant on the support of $\tilde{\mu}$. Thus, within the theorem's assumptions, X_μ improves the stationary-error bound for a fixed biased surrogate.

The threshold $\lambda < \frac{1}{2}$ is a sufficient condition for strict positivity of ν^* . The experiments use $\lambda = 1$ and do not extend the theorem beyond its stated range. The live surrogate also changes with the current flow and is represented by finite batches, so it is not the fixed smooth density assumed here.

References

- Tara Akhound-Sadegh, Jarrod Rector-Brooks, Avishek Joey Bose, Sarthak Mittal, Pablo Lemos, Cheng-Hao Liu, Marcin Sendera, Siamak Ravanbakhsh, Gauthier Gidel, Yoshua Bengio, Nikolay Malkin, and Alexander Tong. Iterated denoising energy matching for sampling from Boltzmann densities. In *International Conference on Machine Learning*, 2024.
- Tara Akhound-Sadegh, Jungyoon Lee, Avishek Joey Bose, Valentin De Bortoli, Arnaud Doucet, Michael Bronstein, Dominique Beaini, Siamak Ravanbakhsh, Kirill Neklyudov, and Alexander Tong. Progressive inference-time annealing of diffusion models for sampling from Boltzmann densities. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

- Michael Samuel Albergo and Eric Vanden-Eijnden. NETS: A non-equilibrium transport sampler. In *International Conference on Machine Learning*, pages 1026–1055. PMLR, 2025. arXiv preprint arXiv:2410.02711.
- Shun-ichi Amari. *Information Geometry and Its Applications*, volume 194 of *Applied Mathematical Sciences*. Springer, 2016.
- Luigi Ambrosio, Nicola Gigli, and Giuseppe Savaré. *Gradient Flows in Metric Spaces and in the Space of Probability Measures*. Lectures in Mathematics ETH Zürich. Birkhäuser, 2 edition, 2008.
- Michael Arbel, Alexander G. D. G. Matthews, and Arnaud Doucet. Annealed flow transport Monte Carlo. In *International Conference on Machine Learning (ICML)*, 2021.
- Dominique Bakry, Ivan Gentil, and Michel Ledoux. *Analysis and Geometry of Markov Diffusion Operators*, volume 348 of *Grundlehren der mathematischen Wissenschaften*. Springer, 2014.
- Martin Bauer, Martins Bruveris, and Peter W. Michor. Uniqueness of the Fisher–Rao metric on the space of smooth densities. *Bulletin of the London Mathematical Society*, 48(3):499–506, 2016.
- Julius Berner, Lorenz Richter, and Karen Ullrich. An optimal control perspective on diffusion-based generative modeling. *Transactions on Machine Learning Research*, 2024.
- David A. Case, H. Metin Aktulga, Kellon Belfon, David S. Cerutti, G. Andrés Cisneros, Vinícius W. D. Cruzeiro, Niel Forouzesh, Timothy J. Giese, Andreas W. Götz, Holger Gohlke, Saeed Izadi, Koushik Kasavajhala, Mehmet C. Kaymak, Edward King, Tom Kurtzman, Taisung S. Lee, Pengfei Li, Junmei Liu, Tyler Luchko, Ray Luo, Madushanka Manathunga, Matheus R. Machado, Huan M. Nguyen, Kurt A. O’Hearn, Alexey V. Onufriev, Fu Pan, Sergio Pantano, Rui Qi, Ali Rahnamoun, Daniel R. Roe, Adrian Roitberg, Celeste Sagui, Stephan Schott-Verdugo, Jian Shen, Carlos L. Simmerling, Nikolai R. Skrynnikov, Jeremy Smith, Jason Swails, Ross C. Walker, Junmei Wang, Haixin Wei, Romain M. Wolf, Xiongwu Wu, Yaxin Xue, Darrin M. York, Si Zhao, and Peter A. Kollman. AmberTools. *Journal of Chemical Information and Modeling*, 63(20):6183–6191, 2023. doi: 10.1021/acs.jcim.3c01153.
- Omar Chehab, Anna Korba, Austin Stromme, and Adrien Vacher. Provable convergence and limitations of geometric tempering for Langevin dynamics. In *International Conference on Learning Representations*, volume 2025, pages 5451–5493, 2025.
- Wendy D. Cornell, Piotr Cieplak, Christopher I. Bayly, Ian R. Gould, Kenneth M. Merz, David M. Ferguson, David C. Spellmeyer, Thomas Fox, James W. Caldwell, and Peter A. Kollman. A second generation force field for the simulation of proteins, nucleic acids, and organic molecules. *Journal of the American Chemical Society*, 117(19):5179–5197, 1995.
- Pierre Del Moral, Arnaud Doucet, and Ajay Jasra. Sequential Monte Carlo samplers. *Journal of the Royal Statistical Society: Series B*, 68(3):411–436, 2006.
- Arnaud Doucet, Nando De Freitas, and Neil Gordon. An introduction to sequential Monte Carlo methods. In *Sequential Monte Carlo methods in practice*, pages 3–14. Springer, 2001.
- Simon Duane, A. D. Kennedy, Brian J. Pendleton, and Duncan Roweth. Hybrid Monte Carlo. *Physics Letters B*, 195(2):216–222, 1987. doi: 10.1016/0370-2693(87)91197-X.
- Conor Durkan, Artur Bekasov, Iain Murray, and George Papamakarios. Neural spline flows. *Advances in neural information processing systems*, 32, 2019.
- Peter Eastman, Jason Swails, John D. Chodera, Robert T. McGibbon, Yutong Zhao, Kyle A. Beauchamp, Lee-Ping Wang, Andrew C. Simmonett, Matthew P. Harrigan, Chaya D. Stern, Rafal P. Wiewiora, Bernard R. Brooks, and Vijay S. Pande. OpenMM 7: rapid development of high performance algorithms for molecular dynamics. *PLOS Computational Biology*, 13(7):e1005659, 2017. doi: 10.1371/journal.pcbi.1005659.
- Víctor Elvira, Luca Martino, and Christian P Robert. Rethinking the effective sample size. *International Statistical Review*, 90(3):525–550, 2022.

- Loris Felardos, Jérôme Hénin, and Guillaume Charpiat. Designing losses for data-free training of normalizing flows on Boltzmann distributions. *arXiv preprint arXiv:2301.05475*, 2023.
- Daan Frenkel and Berend Smit. *Understanding Molecular Simulation: From Algorithms to Applications*, volume 1 of *Computational Science Series*. Academic Press, 2 edition, 2002.
- Marylou Gabrié, Grant M Rotskoff, and Eric Vanden-Eijnden. Adaptive Monte Carlo augmented with normalizing flows. *Proceedings of the National Academy of Sciences*, 119(10):e2109420119, 2022.
- Charles J. Geyer. Markov chain Monte Carlo maximum likelihood. In *Computing Science and Statistics: Proc. 23rd Symposium on the Interface*, pages 156–163, 1991.
- W. K. Hastings. Monte Carlo sampling methods using Markov chains and their applications. *Biometrika*, 57(1):97–109, 1970.
- Aaron Havens, Benjamin Kurt Miller, Bing Yan, Carles Domingo-Enrich, Anuroop Sriram, Brandon Wood, Daniel Levine, Bin Hu, Brandon Amos, Brandon Karrer, Xiang Fu, Guan-Horng Liu, and Ricky T. Q. Chen. Adjoint sampling: highly scalable diffusion samplers via adjoint matching. In *International Conference on Machine Learning (ICML)*, 2025.
- Jiajun He, Yuanqi Du, Francisco Vargas, Dinghuai Zhang, Shreyas Padhy, RuiKang OuYang, Carla Gomes, and José Miguel Hernández-Lobato. No trick, no treat: pursuits and challenges towards simulation-free training of neural samplers. In *ICLR 2025 Workshop on Frontiers in Probabilistic Inference*, 2025.
- William G. Hoover. Canonical dynamics: Equilibrium phase-space distributions. *Physical Review A*, 31(3):1695–1697, 1985. doi: 10.1103/PhysRevA.31.1695.
- Koji Hukushima and Koji Nemoto. Exchange Monte Carlo method and application to spin glass simulations. *Journal of the Physical Society of Japan*, 65(6):1604–1608, 1996.
- Araz Jakalian, David B. Jack, and Christopher I. Bayly. Fast, efficient generation of high-quality atomic charges. AM1-BCC model: II. parameterization and validation. *Journal of Computational Chemistry*, 23(16):1623–1641, 2002. doi: 10.1002/jcc.10128.
- Richard Jordan, David Kinderlehrer, and Felix Otto. The variational formulation of the Fokker–Planck equation. *SIAM Journal on Mathematical Analysis*, 29(1):1–17, 1998.
- Johannes Kästner. Umbrella sampling. *WIREs Computational Molecular Science*, 1(6):932–942, 2011. doi: 10.1002/wcms.66.
- Durk P Kingma and Prafulla Dhariwal. Glow: Generative flow with invertible 1x1 convolutions. *Advances in neural information processing systems*, 31, 2018.
- Leon Klein and Frank Noé. Transferable Boltzmann generators. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- Leon Klein, Andreas Krämer, and Frank Noé. Equivariant flow matching. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- Jonas Köhler, Leon Klein, and Frank Noé. Equivariant flows: exact likelihood generative learning for symmetric densities. In *International Conference on Machine Learning (ICML)*, 2020.
- David P. Landau and Kurt Binder. *A Guide to Monte Carlo Simulations in Statistical Physics*. Cambridge University Press, 4 edition, 2014. doi: 10.1017/CBO9781139696463.
- Paul Langevin. Sur la théorie du mouvement brownien. *Comptes rendus hebdomadaires des séances de l’Académie des sciences*, 146:530–533, 1908.
- Ben Leimkuhler and Charles Matthews. *Molecular Dynamics: With Deterministic and Stochastic Numerical Methods*, volume 39 of *Interdisciplinary Applied Mathematics*. Springer, 2015. doi: 10.1007/978-3-319-16375-8.

- Dong C. Liu and Jorge Nocedal. On the limited memory BFGS method for large scale optimization. *Mathematical Programming*, 45:503–528, 1989. doi: 10.1007/BF01589116.
- Jun S Liu. *Monte Carlo strategies in scientific computing*, volume 10. Springer, 2001.
- Yulong Lu, Jianfeng Lu, and James Nolen. Accelerating Langevin sampling with birth–death. *arXiv preprint arXiv:1905.09863*, 2019.
- Alex Matthews, Michael Arbel, Danilo J. Rezende, and Arnaud Doucet. Continual repeated annealed flow transport Monte Carlo. In *ICML*, 2022.
- Nicholas Metropolis, Arianna W. Rosenbluth, Marshall N. Rosenbluth, Augusta H. Teller, and Edward Teller. Equation of state calculations by fast computing machines. *The Journal of Chemical Physics*, 21(6):1087–1092, 1953.
- Laurence Illing Midgley, Vincent Stimper, Gregor NC Simm, Bernhard Schölkopf, and José Miguel Hernández-Lobato. Flow annealed importance sampling bootstrap. In *NeurIPS 2022 AI for Science: Progress and Promises*, 2022.
- Muhammad Ferjad Naeem, Seong Joon Oh, Youngjung Uh, Yunje Choi, and Jaejun Yoo. Reliable fidelity and diversity metrics for generative models. In *International conference on machine learning*, pages 7176–7185. PMLR, 2020.
- Christian A. Naesseth, Fredrik Lindsten, and David M. Blei. Markovian score climbing: variational inference with $\text{KL}(p||q)$. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- Radford M. Neal. Annealed importance sampling. *Statistics and Computing*, 11(2):125–139, 2001.
- Frank Noé, Simon Olsson, Jonas Köhler, and Hao Wu. Boltzmann generators: Sampling equilibrium states of many-body systems with deep learning. *Science*, 365(6457):eaaw1147, 2019.
- Shūichi Nosé. A unified formulation of the constant temperature molecular dynamics methods. *The Journal of Chemical Physics*, 81(1):511–519, 1984. doi: 10.1063/1.447334.
- Nikolas Nüsken and Lorenz Richter. Solving high-dimensional Hamilton–Jacobi–Bellman PDEs using neural networks: perspectives from controlled diffusions and measures on path space. *Partial Differential Equations and Applications*, 2:48, 2021.
- Alexey Onufriev, Donald Bashford, and David A. Case. Exploring protein native states and large-scale conformational changes with a modified generalized Born model. *Proteins: Structure, Function, and Bioinformatics*, 55(2):383–394, 2004. doi: 10.1002/prot.20033.
- Felix Otto. The geometry of dissipative evolution equations: the porous medium equation. *Communications in Partial Differential Equations*, 26(1–2):101–174, 2001.
- Felix Otto and Cédric Villani. Generalization of an inequality by Talagrand and links with the logarithmic Sobolev inequality. *Journal of Functional Analysis*, 173(2):361–400, 2000.
- George Papamakarios, Theo Pavlakou, and Iain Murray. Masked autoregressive flow for density estimation. *Advances in neural information processing systems*, 30, 2017.
- Danyal Rehman, Oscar Davis, Jiarui Lu, Jian Tang, Michael Bronstein, Yoshua Bengio, Alexander Tong, and Avishek Joey Bose. Efficient regression-based training of normalizing flows for Boltzmann generators. In *International Conference on Learning Representations (ICLR)*, 2026.
- Danilo Rezende and Shakir Mohamed. Variational inference with normalizing flows. In *International conference on machine learning*, pages 1530–1538. PMLR, 2015.
- Lorenz Richter and Julius Berner. Improved sampling via learned diffusions. In *International Conference on Learning Representations (ICLR)*, 2024.

- Christian P. Robert and George Casella. *Monte Carlo Statistical Methods*. Springer Texts in Statistics. Springer, 2 edition, 2004. doi: 10.1007/978-1-4757-4145-2.
- Gareth O. Roberts and Richard L. Tweedie. Exponential convergence of Langevin distributions and their discrete approximations. *Bernoulli*, 2(4):341–363, 1996. doi: 10.2307/3318418.
- Michael Schaefer and Martin Karplus. A comprehensive analytical treatment of continuum electrostatics. *The Journal of Physical Chemistry*, 100(5):1578–1599, 1996. doi: 10.1021/jp9521621.
- Henrik Schopmans and Pascal Friederich. Temperature-annealed Boltzmann generators. In *International Conference on Machine Learning (ICML)*, 2025. arXiv:2501.19077.
- Henrik Schopmans, Christopher von Klitzing, and Pascal Friederich. Efficient training of Boltzmann generators using off-policy log-dispersion regularization. *arXiv preprint arXiv:2602.03729*, 2026.
- John Skilling. Nested sampling for general Bayesian computation. *Bayesian Analysis*, 1(4):833–859, 2006.
- Nikola Surjanovic, Saifuddin Syed, Alexandre Bouchard-Côté, and Trevor Campbell. Parallel tempering with a variational reference. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- Robert H. Swendsen and Jian-Sheng Wang. Replica Monte Carlo simulation of spin-glasses. *Physical Review Letters*, 57(21):2607–2609, 1986.
- Saifuddin Syed, Alexandre Bouchard-Côté, George Deligiannidis, and Arnaud Doucet. Non-reversible parallel tempering: a scalable highly parallel MCMC scheme. *Journal of the Royal Statistical Society Series B*, 84(2):321–350, 2022.
- Charlie B. Tan, Avishek Joey Bose, Chen Lin, Leon Klein, Michael M. Bronstein, and Alexander Tong. Scalable equilibrium sampling with sequential Boltzmann generators. In *International Conference on Machine Learning (ICML)*, 2025.
- Glenn M. Torrie and John P. Valleau. Nonphysical sampling distributions in Monte Carlo free-energy estimation: Umbrella sampling. *Journal of Computational Physics*, 23(2):187–199, 1977. doi: 10.1016/0021-9991(77)90121-8.
- Lorenz Vaitl, Kim A. Nicoli, Shinichi Nakajima, and Pan Kessel. Gradients should stay on path: better estimators of the reverse- and forward-KL divergence for normalizing flows. *Machine Learning: Science and Technology*, 2022.
- Francisco Vargas, Will Grathwohl, and Arnaud Doucet. Denoising diffusion samplers. In *International Conference on Learning Representations*, 2023.
- Cédric Villani. *Optimal Transport: Old and New*, volume 338 of *Grundlehren der mathematischen Wissenschaften*. Springer, 2009.
- Annika von Klitzing, Denis Blessing, Henrik Schopmans, Pascal Friederich, and Gerhard Neumann. Learning Boltzmann generators via constrained mass transport. In *International Conference on Learning Representations (ICLR)*, 2026.
- Martin J. Wainwright and Michael I. Jordan. Graphical models, exponential families, and variational inference. *Foundations and Trends in Machine Learning*, 1(1–2):1–305, 2008.
- Junmei Wang, Romain M. Wolf, James W. Caldwell, Peter A. Kollman, and David A. Case. Development and testing of a general Amber force field. *Journal of Computational Chemistry*, 25(9):1157–1174, 2004. doi: 10.1002/jcc.20035.
- Peter Wirsberger, Andrew J. Ballard, George Papamakarios, Stuart Abercrombie, Sébastien Racanière, Alexander Pritzel, Danilo Jimenez Rezende, and Charles Blundell. Targeted free energy estimation via learned mappings. *The Journal of Chemical Physics*, 153(14), 2020.

- Dongyeop Woo and Sungsoo Ahn. Iterated energy-based flow matching for sampling from Boltzmann densities. In *NeurIPS*, 2024. arXiv:2408.16249.
- Dawn B. Woodard, Scott C. Schmidler, and Mark Huber. Conditions for rapid mixing of parallel and simulated tempering on multimodal distributions. *The Annals of Applied Probability*, 19(2):617–640, 2009.
- Hao Wu, Jonas Köhler, and Frank Noé. Stochastic normalizing flows. *Advances in neural information processing systems*, 33:5933–5944, 2020.
- Leo Zhang, Peter Potapchik, Jiajun He, Yuanqi Du, Arnaud Doucet, Francisco Vargas, Hai-Dang Dau, and Saifuddin Syed. Accelerated parallel tempering via neural transports. *arXiv preprint arXiv:2502.10328*, 2025.
- Qinsheng Zhang and Yongxin Chen. Path integral sampler: A stochastic control approach for sampling. In *International Conference on Learning Representations*, 2022.